



DEBRE TABOR UNIVERSITY
GAFAT INSTITUTE OF TECHNOLOGY
DEPARTMENT OF INFORMATION TECHNOLOGY
TITLE: VITAL EVENT REGISTRATION AND MANAGEMENT SYSTEM
FOR SOUTH GONDAR ZONE.
SUBMITTED AS PARTIAL FULFILLMENT OF THE REQUIREMENTS OF
THE BACHELOR'S DEGREE IN INFORMATION TECHNOLOGY.

Prepared By: Group1

Name	ID
1. Dawit Mengesha.....	0945
2. Fasika Aschale.....	0526
3. Helen Ambaw.....	0350
4. Mekdes Wudu.....	0402
5. Tesfaw Amare.....	1530

Advisor Name : Mr. Simachew

Date: 9/30/2025 G.C

Debre Tabor, Ethiopia

ABSTRACT

The Vital Events Registration and Management System (VERMS) is a web-based information system designed to modernize and automate the manual registration of vital events birth, death, marriage, and divorce in the South Gondar Zone, Ethiopia. The existing paper-based process suffers from inefficiencies, delays, data errors, and security risks. This project proposes a digital solution that replaces traditional workflows with a secure, centralized database and responsive web interfaces accessible at Kebele, Woreda, and Zone administrative levels.

Using the Iterative Waterfall Software Development Model, the system was designed with HTML, CSS, and JavaScript for the front end, PHP for the back end, and MySQL for the database. Key features include automated data validation, role-based access control, QR-coded digital certificates, and real-time reporting dashboards. UML diagrams (Use Case, Sequence, Activity, and Class) were developed to model system behavior and architecture.

The system aims to reduce registration time, improve data accuracy, and enable evidence-based decision-making through timely demographic reports. A feasibility analysis confirmed the system's operational, technical, legal, and economic viability. Overall, VERMS demonstrates a scalable and practical model for digital transformation in vital events registration, aligned with Ethiopia's Digital Transformation Strategy 2025.

ACKNOWLEDGEMENTS

First and foremost, we give thanks to Almighty God for granting us the strength, wisdom, and perseverance to complete this project successfully.

We would also like to express our deepest gratitude to Debre Tabor University, particularly the Gafat Institute of Technology, Department of Information Technology, for providing us with the academic foundation, resources, and guidance necessary to complete this project successfully.

Our sincere appreciation goes to our project advisor, Mr. Simachew, for his continuous support, constructive feedback, and invaluable insights throughout all stages of this work. His encouragement and technical advice were instrumental in shaping the direction and quality of this project.

Finally, we extend heartfelt thank the South Gondar Zone Vital Events Registration Office for granting us access to valuable information, operational insights, and practical feedback that informed the system's design. Special thanks to the Kebele, Woreda, and Zone staff who participated in interviews and observations during the data collection phase.

TABLE of CONTENTS

ABSTRACT	i
ACKNOWLEDGEMENTS	ii
TABLE of CONTENTS	iii
LIST OF ABBREVIATION	vii
CHAPTER 1: INTRODUCTION	1
1.1 Background (the Organization and the Project)	1
1.2 Motivation	2
1.3 Statement of the Problem	3
1.4 Objective of the Project	4
1.4.1 General Objective	4
1.4.2 Specific Objectives	4
1.5 Scope of the Project	5
1.6 Significance of the Project	6
1.7 Feasibility Study	7
1.7.1 Operational Feasibility	7
1.7.2 Economic Feasibility	7
1.7.3 Legal Feasibility	8
1.7.4 Technical Feasibility	8
1.8 Methodology	9
1.8.1 Data Collection Method	9
1.8.2 Software Process Model	10
1.8.3 Tools and Techniques	10
1.9 Roles and Responsibilities	11
1.10 Work Breakdown Structure	12
1.11 Schedule	13
CHAPTER 2: STUDY OF EXISTING SYSTEM	14
2.1 Introduction	14
2.2 Existing System	14
2.3 Organization Structure	15
2.4 Services Provided	16
2.5 Users	16
2.6 Business Rule Identification	17
2.7 Existing Infrastructure	17
2.8 Proposed System	19
CHAPTER 3: SOFTWARE REQUIREMENT SPECIFICATION	23
3.1 Introduction	23
3.2 General Constraints	23
3.3 Specific Requirements	23
3.3.1 External Interface Requirements	23
3.4 Functional Requirements	25
3.5 Use Case Design	26
3.5.1 Actor Identifications	26
3.5.2 Use Case Identification	27
3.5.3 Use Case Diagram	28
3.5.4 Use Case Description	28
3.6 Sequence Diagram	36

3.7 Activity Diagram	39
3.8 Class Diagram	42
3.9 Non-Functional Requirements	43
CHAPTER 4: SYSTEM DESIGN	46
4.1 Design Overview	46
4.2 Chosen System Architecture	47
4.3. User Interface Design	48
4.3.1. Description of the User Interface	48
4.3.2. Screenshot Images	49
4.4 Database Design	49
4.4.1. Persistent (ER) Diagram	50
4.4.4. Normalization	50
CHAPTER 5: TESTING	51
5.1 Test Plan	51
5.1.1 Responsibilities of each Group Member	51
5.1.2 Types of Testing	51
5.1.3 Features to be Tested	52
5.1.4 Features not to be Tested	52
5.1.5 Testing Tools	52
5.2 Test Cases	53
CHAPTER 6: CONCLUSION AND RECOMMENDATION	58
6.1 Conclusion	58
6.2 Recommendation	58
6.3 Future Work	59
REFERENCE	61
APPENDICES	62
Appendices A: System Screenshots Login Forms	62
Appendices B: Persistent ER Diagram	62
Appendices C:Source Code Samples	62

LIST OF FIGURE

Figure 3.1: Use Case Diagram	28
Figure 3.2: Sequence Diagram for Login	37
Figure 3.3: Sequence Diagram for Registering Events	37
Figure 3.4: Sequence Diagram for Updating Events	38
Figure 3.5: Sequence Diagram for Create User Account	38
Figure 3.6: Sequence Diagram for Make Payment	39
Figure 3.9: Activity Diagram for Register Events	40
Figure 3.10: Activity Diagram For Update Event	41
Figure 3.11: Activity Diagram for Create Account	42
Figure 3.12: Class Diagram	43
Figure 4.13: Detailed System Architecture Diagram	47
Figure 4.14 Home Page	49
Figure 4.15 Admin Page	49
Figure 4.16: Persistent ER Diagram	50

LIST OF TABLE

Table 1.1: Roles and Responsibilities	11
Table 1.2: Gantt Chart	13
Table 3.3: Use Case Identification	27
Table 3.4: Use Case Description and Actors	28
Table 3.5: Use Case Description of Register Events	30
Table 3.6: Use Case Description for Update Event	31
Table 3.7: Use Case Description for give Printed Certificate	33
Table 3.8: Use Case Description for Generate Report	34
Table 3.9: Use Case Description for Login	35
Table 5.10: Responsibilities of each Group Member	51
Table 5.11: Testing Tools	52

LIST OF ABBREVIATION

Abbreviation	Full Form
API.....	Application Programming Interface
CRVS.....	Civil Registration and Vital Statistics
CSS.....	Cascading Style Sheets
BR.....	Business Rule
ER.....	Entity-Relationship
ETB.....	Ethiopian Birr
HTML.....	HyperText Markup Language
ID.....	Identifier
OWASP.....	Open Worldwide Application Security Project
PHP.....	Hypertext Preprocessor
QR.....	Quick Response (Code)
RAM.....	Random Access Memory
SMS.....	Short Message Service
SQL.....	Structured Query Language
SRS.....	Software Requirements Specification
UI.....	User Interface
UML.....	Unified Modeling Language
UN.....	United Nations
UAT.....	User Acceptance Testing
VERMS.....	Vital Events Registration Management System
ZAP.....	Zed Attack Proxy (security testing tool)

CHAPTER 1: INTRODUCTION

1.1 Background (the Organization and the Project)

Civil registration of vital events is a fundamental function of governance, providing the legal foundation for individual identity and the primary data for public policy. The practice of recording births and other life events has deep historical roots in Ethiopia, with traditions dating back centuries, particularly within the Orthodox Church. In the modern administrative structure, the South Gondar Zone Vital Events Registration Office is the governing body responsible for this critical task, operating through a three-tiered hierarchy of Kebele (local), Woreda (district), and Zone (regional) offices. Despite its importance, the current system remains almost entirely manual and paper-based. This reliance on physical ledgers and carbon-copy forms for registering the four critical events birth, death, marriage, and divorce has created significant bottlenecks. The existing infrastructure is strained, leading to delays, errors, and inefficiencies that directly impact citizens' access to their rights and services.

This project, developed by a team of Information Technology students, proposes the design and implementation of an Automated Vital Events Registration and Management System (VERMS). This digital solution aims to transform the Office's operations by:

1. Replacing error-prone manual workflows with a secure, cloud-based central database to ensure data integrity and permanence.
2. Enhancing real-time data accessibility for Kebele and Woreda offices, eliminating the dependency on slow physical document transfers.
3. Supporting faster legal, administrative, and policy-making processes by providing accurate, immediately available statistical records.

By modernizing this historically significant practice, the project directly aligns with the South Gondar Zone's broader development goals and Ethiopia's national Digital Transformation Strategy aiming to build a more efficient, transparent, and citizen-centric governance model [1] [2].

1.2 Motivation

The development of the Automated Vital Events Registration and Management System (VERMS) is driven by a critical need to address the profound limitations of the existing manual system. Each motivating factor is directly linked to tangible benefits for the primary users: the citizens of South Gondar and the government staff who serve them.

1. To Eliminate Crippling Inefficiencies and Data Errors. The current paper-based process is slow, cumbersome, and prone to human error, with an estimated 18% of forms containing omissions or mistakes. This directly harms citizens through multi-week delays in receiving certificates, which are required for access to education, healthcare, and legal services. VERMS will automate data entry with validation checks, reducing errors processing time from weeks to hours, thereby ensuring citizens can access their rights without unnecessary delay.
2. To Overcome the Barrier of Data Inaccessibility: Kebele and Woreda offices currently operate as informational silos, unable to query the central database at the Zone office. This forces a reliance on the physical transport of paper documents, a process that can take weeks and often results in incomplete data. VERMS will provide real-time, role-based access to a centralized database for all authorized offices, enabling instant verification of records, improving inter-office collaboration, and ensuring that decisions are made using complete and current information.
3. To Enable Data-Driven Governance and Policy Formulation: Accurate and timely vital statistics are the bedrock of effective public policy, planning, and resource allocation [3]. The manual system's lag in data compilation renders it nearly useless for responsive governance. VERMS will automate the generation of accurate demographic reports and statistical analyses, providing zone administrators with the real-time data needed to make evidence-based decisions in critical areas like public health, education planning, and infrastructure development.
4. To Drastically Reduce Operational Costs and Resource Waste: The existing system incurs significant and recurring expenses for paper, printing, storage, and the transportation of documents between offices. VERMS will eliminate over 80% of these physical resource costs by digitizing the entire workflow. The

freed-up budget and resources can then be reallocated to improve other public services, directly benefiting the community.

5. **To Ensure Security, Preservation, and Confidentiality:** Paper records are highly vulnerable to damage from environmental factors (e.g., moisture, pests), loss, theft, and unauthorized viewing, posing a significant risk to citizen privacy. VERMS will safeguard sensitive citizen data through encrypted digital storage, secure access controls, and automated audit trails, ensuring the long-term preservation and confidentiality of vital records for generations to come.

This project is motivated by the urgent opportunity to leverage technology not merely for automation, but to fundamentally enhance service delivery, empower government officials with better tools, and restore citizens' trust in a critical public institution.

1.3 Statement of the Problem

The South Gondar Zone Vital Events Registration Office is hindered by a fundamentally outdated, manual, and paper-based system for recording vital events. This core inadequacy manifests in three critical and interconnected problems that cripple efficiency, compromise data integrity, and undermine public service delivery:

1. **Protracted and Inefficient Service Delivery:** The absolute reliance on physical paperwork necessitates a slow, multi-stage process where forms must be manually transported from Kebele to Woreda to Zone offices. This logistical bottleneck creates extensive delays to receive essential certificates. This directly impedes their ability to access basic rights and services, such as enrolling children in school, claiming inheritance, or receiving healthcare benefits, in a timely manner.
2. **Poor Data Integrity and Isolated Information Silos:** The manual data entry process is highly susceptible to human error, with approximately 18% of forms containing omissions, illegible information, or inaccuracies. Furthermore, the inability of Kebele and Woreda offices to access the central database creates severe information silos. Staff cannot instantly verify existing records or prevent duplicates, and the data available for daily operations and decision-making is often incomplete, outdated, and unreliable [4].
3. **Exorbitant Operational Costs and Significant Security Risks:** The system incurs substantial and recurring financial costs for paper, printing, storage infrastructure, and the transportation of documents. More critically, the physical nature of the

records makes them highly vulnerable to damage from environmental factors (e.g., fire, water, pests), loss, theft, and unauthorized access. This poses a severe and continuous risk to the confidentiality and long-term preservation of sensitive citizen information.

These problems collectively result in a failure to provide a modern, efficient, and secure service, eroding public trust and hindering the zone's administrative and developmental objectives. The transition to a digital system is not merely an operational improvement but a necessary modernization to fulfill the office's core mandate effectively [4].

1.4 Objective of the Project

1.4.1 General Objective

To design and implement an automated vital events registration system for the South Gondar Zone.

1.4.2 Specific Objectives

To achieve the general objective, the following specific, measurable, achievable, relevant, and time-bound (SMART) objectives are defined:

- ✓ To automate and streamline the registration process for birth, death, marriage, and divorce events, reducing the average time to complete a registration.
- ✓ To minimize data loss and entry errors by implementing digital form validation and centralized cloud storage, achieving a data accuracy and reducing form-based errors.
- ✓ To enable real-time, secure access for Kebele and Woreda offices through a responsive, web-based interface, ensuring system availability of operational hours and allowing authorized users to retrieve any record.
- ✓ To automate the generation and issuance of official, tamper-evident certificates (e.g., birth, death) and standard administrative reports, enabling the production of any certificate.
- ✓ To support robust statistical analysis and reporting for zone-level planning by integrating data visualization and export features, allowing administrators to generate comprehensive demographic reports (e.g., population trends, birth rates by Woreda) with a single click.

1.5 Scope of the Project

This section defines the boundaries of the Vital Events Registration and Management System (VERMS), explicitly outlining what functionalities, technologies, and user groups will be included, as well as what will be explicitly excluded from this phase of the project.

In-Scope

1. Functional Scope:
 - a. The digitization of the complete registration workflow for the four core vital events: birth, death, marriage, and divorce.
 - b. Serving three primary tiers of users with role-specific interfaces and permissions: Kebele (data entry and initial verification), Woreda (validation and quality control), and Zone (final approval, certification, and reporting) offices.
 - c. Core features including citizen registration, record management (create, read, update), certificate generation, user access control, and basic reporting.
 - d. Update functionality will require appropriate administrative permissions and audit trails, as per business rules.
2. Technical Scope:
 - a. Frontend: A responsive web application built with HTML5, CSS3, and JavaScript to ensure a consistent user experience across different devices and screen sizes.
 - b. Backend: A robust server-side application developed in PHP to handle business logic, with a MySQL relational database for secure and structured data storage.
 - c. Network: The system will be designed to operate over existing LAN/WAN infrastructure, with considerations for secure access and data synchronization across geographically dispersed offices.
 - d. Security: Implementation of foundational security measures, including role-based access control (RBAC), password hashing, and input validation.
3. User Scope:
 - a. Government employees at Kebele, Woreda, and Zone-level vital events offices.
 - b. System administrators responsible for user management and system maintenance.

- c. Citizens are the beneficiaries of the service but will interact with the system through government officials in this phase.

1.6 Significance of the Project

The successful implementation of the VERMS will deliver immediate operational improvements and generate significant long-term strategic benefits for all stakeholders, fundamentally enhancing governance and service delivery in the South Gondar Zone.

1. For Government Employees (Kebele, Woreda, and Zone Staff): The system will transform their daily work experience by eliminating tedious manual paperwork, reducing frustrating errors, and providing instant access to information. This will lead to higher job satisfaction, allow them to serve more citizens effectively, and enable a shift from administrative data handlers to knowledgeable service advisors. Long-term, this builds a more skilled, efficient, and motivated civil service.
2. For the South Gondar Zone Administration: VERMS moves the administration from relying on outdated, estimated data to making decisions based on accurate, real-time demographic intelligence. This is crucial for effective resource allocation, strategic planning in healthcare and education, and designing targeted policies (e.g., to address regional health trends or manage school placements). Long-term, this fosters a culture of evidence-based governance, increasing the efficacy of public programs and building trust in public institutions.
3. For Citizens and Communities: This project directly translates to tangible improvements in quality of life. Families will no longer face debilitating delays in obtaining birth certificates for school enrollment or death certificates for settling affairs. The reduction in bureaucratic hurdles empowers citizens, especially the poor and marginalized, to access their rights and public services with dignity and speed. Long-term, a reliable civil registration system is a cornerstone of social inclusion and economic participation, ensuring every individual is recognized by the state.
4. For the Judiciary and Legal System: Courts will benefit from the ability to quickly and reliably verify vital events, directly reducing delays in legal proceedings related to inheritance, child custody, and marital disputes. The system's secure

and audit-able records provide a trusted source of evidence, strengthening the integrity of the legal process. Long-term, this enhances the pace and fairness of justice.

5. For National Development: This project serves as a scalable proof-of-concept for digital transformation within the Ethiopian public sector. The lessons learned, and the platform developed, can be adapted and replicated by other zones and regions, contributing directly to the national goals outlined in Ethiopia's Digital Transformation Strategy 2025. Long-term, it strengthens the entire country's civil registration infrastructure, leading to better national statistics and planning.

In essence, this project transcends mere technical automation; it is an investment in more transparent, efficient, and equitable governance that will yield dividends for the South Gondar Zone for years to come.

1.7 Feasibility Study

A comprehensive feasibility study was conducted to determine the practicality and viability of the proposed VERMS. The analysis confirms that the project is feasible across all critical dimensions.

1.7.1 Operational Feasibility

The project is highly feasible from an operational perspective. The intended users have demonstrated a strong willingness to adopt the new system.

1. Management Support: The South Gondar Zone Vital Events Registration Office management has provided a written commitment of support, agreeing to allocate time for staff training and to champion the adoption process. This is driven by their acute awareness of the current system's crippling inefficiencies.
2. User Acceptance and Training: The user-friendly, web-based interface is designed with low computer literacy in mind, featuring intuitive forms and clear Amharic language labels. A staggered training plan has been developed, requiring initial training for Kebele registrars and Woreda and Zone administrators, minimizing disruption to daily operations.
3. Organizational Impact: The system aligns perfectly with the existing workflow and hierarchical approval structure, ensuring a smooth integration into daily routines without requiring a fundamental restructuring of staff responsibilities.

1.7.2 Economic Feasibility

The system presents a strong economic case with a rapid return on investment (ROI) through significant cost savings and efficiency gains.

➤ **Tangible Benefits for Cost Savings:**

Printing and Paper: The system eliminates recurring expenses previously spent on forms, carbon paper, and stationery through complete digitization of registration processes and certificate issuance.

Transportation: The need for physically transferring documents between offices is removed, leading to considerable reductions in fuel and courier service expenses.

Storage: The reliance on physical archives is minimized, reducing long-term costs associated with maintaining and expanding storage facilities.

➤ **Intangible Benefits & ROI:**

The value of time savings is substantial. Reducing the average registration time translates into faster service delivery and improved citizen satisfaction.

The ability to make timely, data-driven decisions also has a significant, though harder to quantify, positive economic impact on public spending.

1.7.3 Legal Feasibility

The system has been designed to be fully compliant with Ethiopian law.

- ✓ **Data Protection:** The architecture and data handling protocols adhere to Ethiopia's Data Protection Proclamation (No. 1321/2024), ensuring citizen data is collected, stored, and processed lawfully.
- ✓ **Business Rules:** All existing legal and administrative business rules governing vital events registration (e.g., witness requirements for marriage, court orders for amendments) are hardcoded into the system's logic, ensuring full compliance.
- ✓ **Evidentiary Standards:** The system's secure audit trails and tamper-evident digital certificates are designed to meet the evidentiary standards required for admissibility in court proceedings.

1.7.4 Technical Feasibility

This is a study of resource availability that may affect the ability to achieve an acceptable system. This evaluation determines whether the technology needed for the proposed system is available or not.

- ✓ Can the work for the project be done with current equipment, existing software technology, and available staff?
- ✓ Can the system be implemented after development?
- ✓ If new technology is needed, then what can be developed?

This is concerned with specifying equipment and software that will successfully satisfy the user requirements. An important issue for the development of a project is the selection of suitable front-end and back-end.

➤ Front-end selection:

- ✓ It must have a graphical user interface that assists employees who do not have IT knowledge.
 - ✓ Must be according to the organization's requirements and business rules.
 - ✓ Must provide excellent reporting features with good printing support.
 - ✓ Platform-independent.
 - ✓ Easy to debug and maintain.
 - ✓ Front-end must integrate with a back-end like PHP and a database like MySQL.
- According to the above-stated features, we select JavaScript, HTML, and CSS as the front-end for developing our project.

➤ Back-end selection:

- ✓ Multiple user support.
- ✓ Efficient data handling.
- ✓ Provide an efficient security system.
- ✓ Efficient data retrieval and maintenance.
- ✓ Operating system compatible.
- ✓ Easy to install.
- ✓ Easy to debug and maintain.
- ✓ Easy to connect with the front-end.

The selection of JavaScript, HTML, and CSS for the front-end and MySQL and PHP for the back-end is justified by their alignment with the project's requirements for a user-friendly, secure, and scalable vital event registration system. These technologies are mature, widely supported, and capable of leveraging existing equipment and staff skills. They ensure a platform-independent, maintainable solution with robust reporting and data management capabilities, making them the optimal choice for the project's technical feasibility.

1.8 Methodology

1.8.1 Data Collection Method

A mixed-methods approach will be used to gather comprehensive requirements and understand the operational context.

➤ Primary Sources:

- ✓ Semi-Structured Interviews: Conducted across Kebele, Woreda, and Zone offices, including registrars, data clerks, and administrators. An interview protocol will be used to ensure consistency while allowing for exploratory discussion.
- ✓ Direct Observation: The project team will spend time at registration offices to map the complete as-is process workflow, document pain points, time duration's, and exception handling not captured in interviews.
- ✓ Document Analysis: Existing paper forms, logbooks, report templates, and operational manuals will be analyzed to extract business rules, data fields, and validation criteria [5].

Secondary Sources: Internet research for system samples and tutorials.

1.8.2 Software Process Model

The Iterative Waterfall Model has been selected for this project. This model provides the necessary structure for a well-defined project with clear requirements while allowing for feedback between phases [5].

The sequential phases will be:

1. Requirements Analysis & Specification
2. System Design (Architecture, UI/UX, Database)
3. Implementation (Coding)
4. Testing (Unit, Integration, System, User Acceptance)
5. Deployment & Maintenance

At the end of each phase, a formal review with the project advisor and stakeholders will be conducted before proceeding to the next phase. This ensures disciplined progress and early detection of errors.

1.8.3 Tools and Techniques

A modern and robust technology stack will be used to ensure the development of a secure, maintainable, and efficient system.

- ✓ Frontend Development: HTML5, JavaScript and CSS3 to build a dynamic, responsive, and single-page application (SPA) that delivers a desktop-like user experience.
- ✓ Backend Development: PHP to create a secure, scalable, and well-structured application following the Model-View-Controller (MVC) architectural pattern.

- ✓ Database Management: MySQL for robust, relational data storage, ensuring data integrity and supporting complex queries for reporting.
- ✓ Modeling & Design: EdrawMax, Lucidchart and Draw.io will be used to create all UML diagrams (Use Case, Sequence, Activity, Class Diagrams) for clear design communication.
- ✓ Testing: A multi-faceted testing strategy will be employed:
 1. Functional Testing: Selenium for automated browser testing (Firefox, Chrome).
 2. Performance Testing: Apache JMeter to ensure the system can handle concurrent user loads.
 3. Security Testing: Manual code reviews and OWASP ZAP for vulnerability scanning.
 4. User Acceptance Testing (UAT): Real-world testing by selected Zone office staff.
- ✓ Version Control: Git with GitHub will be used for source code management, enabling collaboration and tracking all changes throughout the project life cycle.

1.9 Roles and Responsibilities

Table 1.1: Roles and Responsibilities

Team Member	Role	Responsibilities
Tesfaw	Project Manager/System Analyst	✓ Oversee project progress, coordinate tasks, ensure deadlines, gather and document requirements, and design UML diagrams (use case, sequence, activity, class).
Dawit	Lead Developer	✓ Code front-end (HTML, CSS, JavaScript) and back-end (PHP) components, integrate with MySQL database, and ensure code quality.
Mekdes	Database Designer	✓ Design and normalize database schema, ensure data integrity, and manage database connectivity with the application.
Helen	Tester/Quality Assurance	✓ Develop test plans, conduct unit, integration, usability, and performance testing, and validate system functionality.

Fasika	Security Specialist/Training Coordinator	✓ Implement security features (authentication, session management, password encryption), conduct security testing, develop training materials, and conduct user training sessions in Amharic and English.
--------	--	---

1.10 Work Breakdown Structure

The project work is decomposed into manageable phases and tasks. A Table (Table 1.2) provides a visual timeline for this plan.

Phase 1: Requirements & Planning (1 month)

Task 1.1: Conduct stakeholder interviews and observations

Task 1.2: Document Software Requirements Specification (SRS)

Task 1.3: Develop UML models (Use Cases, Class, Activity Diagrams)

Task 1.4: Create project schedule and finalize budget

Phase 2: System Design (1 month)

Task 2.1: Design database schema and architecture

Task 2.2: Create wire frames and UI mockups

Task 2.3: Select and set up development tools and environment

Phase 3: Implementation (2 month)

Task 3.1: Develop core database and backend modules

Task 3.2: Implement frontend user interfaces

Task 3.3: Integrate frontend and backend components

Task 3.4: Implement user authentication and security features

Phase 4: Testing & Deployment (1 month)

Task 4.1: Execute unit, integration, and system testing

Task 4.2: Conduct User Acceptance Testing (UAT) with zone staff

Task 4.3: Deploy system to a production-like environment

Task 4.4: Develop user manuals and conduct training

Phase 5: Project Closure (1 month)

Task 5.1: Prepare final project documentation and source code

Task 5.2: Deliver final presentation and project defense

1.11 Schedule

Table 1.2: Gantt Chart

Task	Duration In Months					
	1	2	3	4	5	6
Requirements and Planning						
System Design						
Implementation						
Testing and Deployment						
Project Closure						

CHAPTER 2: STUDY OF EXISTING SYSTEM

2.1 Introduction

A thorough analysis of the existing system is a critical prerequisite for successful digital transformation. This chapter provides a comprehensive examination of the manual vital events registration process within the South Gondar Zone. The purpose of this study is to deconstruct the current workflows, identify systemic inefficiencies, and understand user pain points. This foundational understanding ensures that the proposed automated system is designed to directly address the root causes of existing problems, rather than merely superficially automating a flawed process.

2.2 Existing System

The South Gondar Zone's vital events registration system currently operates through a completely manual, paper-based workflow that has remained fundamentally unchanged for decades. This labor-intensive process begins at the Kebele level, where local managers assisted by community nominators first record vital events births, deaths, marriages, and divorces on standardized paper forms. Each event type follows specific protocols; for instance, marriage registration requires four witnesses (two representing each spouse) and three passport-sized photographs, while birth registration demands the physical presence of parents or a court-appointed guardian.

The system's architecture follows a strict hierarchical model. After initial recording at Kebele offices, carbon-copied forms travel through multiple administrative layers first to Woreda district offices, then finally to the Zone's central registry. This physical document transfer creates significant bottlenecks in data processing.

Several structural limitations plague the current system. First, the complete reliance on paper records makes the data vulnerable to both environmental damage and human error. Zone office reports indicate physical records sustain damage annually from moisture, pests, or improper handling. Second, the absence of digital validation mechanisms leads to frequent data quality issues submitted forms contain errors or omissions that require corrective follow-up. Third, the system's fragmented nature prevents real-time data access; Kebele and Woreda offices cannot verify records against the Zone's central registry without initiating time-consuming manual searches.

The registration process imposes notable burdens on citizens. Applicants must typically make multiple trips to government offices. Fee structures vary by event type, ranging from 10 ETB for death certificates to 35 ETB for birth and marriage registrations. These financial and time costs disproportionately affect rural populations, as the system provides no remote submission options.

From an administrative perspective, the manual system severely limits analytical capabilities. The Central Statistics Agency struggles to perform timely demographic analysis, while courts face difficulties retrieving historical records for legal proceedings.

Security concerns further complicate operations. The paper-based system lacks robust access controls, creating risks of unauthorized record tampering. No centralized audit trail exists to track document modifications or accesses. Perhaps most critically, the system cannot prevent duplicate registrations.

This exhaustive examination of the existing system's limitations directly informs our proposed automated solution's design parameters, particularly regarding data integrity safeguards, processing efficiency improvements, and accessibility enhancements for both civil servants and citizens. The subsequent sections will demonstrate how digital transformation can address these documented pain points while maintaining compliance with all legal and operational requirements.

2.3 Organization Structure

The South Gondar Zone's vital events registration system operates through a clearly defined three-tier administrative hierarchy at the foundation, 47 Kebele (local) offices serve as primary data collection points, where trained registrars manually record events in standardized ledgers. These local offices report to 12 Woreda (district) offices that function as intermediary processing centers, responsible for form verification and temporary archival. The Zone's central registry office in Debre Tabor serves as the ultimate authority, maintaining master records and issuing certified documents.

Staffing follows a pyramid distribution model, with Kebele offices typically employing a small number of registrars, Woreda offices maintaining medium-sized teams, and the Zone office operating with a larger group of permanent staff members supplemented by seasonal contractors during peak registration periods. This structure creates several operational challenges.

Knowledge Disparities: Training quality varies significantly across levels. A lower proportion of Kebele staff receive annual refresher courses compared to the higher coverage achieved at the Zone level, resulting in uneven capacity and performance.

Resource Allocation: Document storage capacity remains uneven, with Kebele, Woreda and Zone offices.

Communication Gaps: The hierarchical reporting model creates information latency.

The organizational workflow mandates strict separation of duties:

- ✓ Kebele Registrars: Verify citizen identities and event authenticity
- ✓ Woreda Clerks: Perform data quality checks and temporary archiving
- ✓ Zone Officers: Authorize certificates and compile statistics

This division of labor, while theoretically sound, suffers from inconsistent implementation.

2.4 Services Provided

The South Gondar Zone Vital Events Registration Office delivers four core services through its manual operational framework. These services constitute the fundamental interface between civil administration and citizens for legal documentation needs.

➤ Registration of Vital Events:

The office maintains jurisdiction over recording four critical life events:

1. **Birth Registration:** Requires presentation of both parents' Kebele ID cards and physical attendance at Kebele offices. In cases of parental absence, a court-issued delegation document becomes mandatory.
2. **Death Certification:** Demands submission of a medical death certificate from authorized health centers alongside next-of-kin verification.
3. **Marriage Registration:** Involves a multi-step process requiring four witnesses (two per spouse), three passport photographs, and biometric verification of all parties.
4. **Divorce Documentation:** Necessitates production of original marriage certificates and court dissolution orders.

This service ecosystem demonstrates both the critical importance of vital registration and the systemic inefficiencies plaguing manual operations. The documented service delays and quality issues directly motivate the proposed digital transformation outlined in subsequent chapters.

2.5 Users

- ✓ Primary Users: South Gondar Zone Vital Events Registration Office staff (registrars, clerks, administrators).
- ✓ Secondary Users: Citizens requesting services.
- ✓ External Users: Zone administration, Central Statistics Agency (zone level), courts, health centers, schools.

2.6 Business Rule Identification

Business rules are definitive, specific, and testable directives that define or constrain aspects of business operations. They are derived from laws, regulations, organizational policies, and standard procedures. Their purpose is to assert business structure, control behavior, and ensure the integrity and consistency of business processes. For the VERMS, these rules are paramount to ensuring the legal validity, security, and reliability of the vital registration process.

The following core business rules have been identified, categorized, and specified for implementation within the system:

BR1: Any vital event cannot be registered more than once.

BR2: The person who wants to register the vital event must appear physically in the Kebele registrar offices.

BR3: The vital event information must be registered on the forms without any error.

BR4: The registrars update events with the order of the courts.

BR5: The registrar must assure the correctness of the registered information by showing it to the users.

BR6: After reading the registered information, the user must sign on it.

BR7: The vital events should be registered with the working language of English and Amharic.

BR8: The registrar must also sign on registration papers after assuring their correctness.

BR9: The certificate is given to the user after they pay the service payment.

BR10: The registrar should keep the confidentiality of the vital event information.

BR11: Every employee in every stage accesses the system according to their privilege.

2.7 Existing Infrastructure

The existing manual system operates within severe technological constraints that fundamentally limit its effectiveness. These infrastructure challenges permeate every level of the registration process, creating systemic bottlenecks that no amount of procedural refinement can overcome.

- **Physical Infrastructure**

Document Storage Systems: Kebele offices rely on outdated steel cabinets that are highly vulnerable to moisture and pests. Woreda offices have very limited records storage space, often insufficient for long-term retention. At the Zone headquarters, critical overcrowding is common, with stacks of ledgers spilling into walkways and work areas, severely affecting both safety and efficiency.

Transportation Networks: Most transfers from Kebele to Woreda offices depend on informal couriers. During the rainy season, poor road conditions frequently cause significant delays in document shipments. There is no dedicated budget for document transport, forcing offices to reallocate funds from other operational areas to cover these expenses.
- **Technological Gaps**

Communication Systems: Only a small fraction of Kebele offices have access to reliable landline services, and internet connectivity remains very low, particularly at the Woreda level. As a result, official communications still rely heavily on paper memos, which can take several days to reach their destination, creating delays in decision-making and coordination.

Equipment Shortages: Manual typewriters, many of which are decades old, continue to serve as the primary data entry tools. Chronic shortages of carbon paper force registrars to manually copy forms multiple times. Only a limited number of offices have functioning calculators, making basic statistical work slow and prone to error.
- **Data Management Challenges**

Information Retrieval: Locating specific records requires time-consuming manual searches through chronological ledgers. Retrieving recent records takes considerable time, while accessing older archival records can be significantly longer. There is no cross-referencing system to connect related events, such as linking birth records to marriage records.

Backup Vulnerabilities: Each administrative level faces single-point failure risks, with no formal disaster recovery protocols in place for damaged or lost records. Historical documents also show steady physical degradation over time, increasing the risk of permanent information loss.
- **Human Resource Constraints**

Training Deficits: Only a minority of staff members receive annual refresher training, and there is no standardized competency framework for registrars.

High staff turnover at the Kebele level further weakens institutional memory and consistency in service delivery.

Workflow Inefficiencies:A large share of staff time is spent on manual form handling rather than value-added tasks. During peak registration periods, offices are forced to rely heavily on overtime to manage backlogs. Ergonomic considerations for staff performing repetitive tasks are generally lacking.

➤ **Financial Realities**

Budget Allocation:The majority of operational funds are consumed by paper and printing costs, with only a small portion allocated to technological investments. There is no capital budget dedicated to improving infrastructure, resulting in continued reliance on outdated systems.

Cost of Continuity:Ongoing expenses for transporting documents are substantial, and storage costs continue to increase each year. Additionally, the financial impact of errors and inefficiencies within the system represents a significant hidden cost that strains the limited budget available.

These infrastructure limitations create a self-reinforcing cycle of inefficiency. Outdated technology increases processing times, which drives up backlog volumes, which further strains physical storage capacity. The proposed digital solution directly addresses these pain points through:

1. Cloud-based storage eliminating physical space constraints
2. Mobile data entry reducing transportation dependencies
3. Automated workflows optimizing staff utilization
4. Distributed access overcoming communication barriers

The subsequent technical design chapters will demonstrate how modern information systems can overcome these legacy constraints while maintaining the rigorous verification standards that underpin public trust in vital registration systems.

2.8 Proposed System

The existing manual system, while foundational, is fundamentally incapable of scaling to meet the demands of South Gondar Zone's population. Characterized by certificate issuance delays and a data error, it acts as a bottleneck to development and service delivery. In direct response, we propose the Vital Events Registration and Management System (VERMS) a comprehensive digital solution designed to

modernize operations while respecting and encoding the zone's established legal and procedural frameworks.

2.8.1 Core Architectural Shift: from Paper to Digital

VERMS initiates a fundamental architectural shift by replacing vulnerable paper-based workflows with a secure, centralized digital infrastructure.

1. **Efficiency and Accessibility:** The system introduces digital forms with built-in validation, drastically reducing errors and omissions. Citizens will have multiple submission channels: online via a secure portal, at designated community kiosks, or through trained registrars. For the 90% of Kebele offices with unreliable internet, registrars will use offline-capable tablets and mobile devices. Data captured offline will be automatically synchronized with the central database the moment a connection is re-established, ensuring no community is left behind.
2. **Real-Time Data Access:** A single, cloud-based database will replace fragmented paper files. This grants Kebele, Woreda, and Zone offices real-time, role-based access to the same information. A registrar in a remote village can instantly verify if a birth has already been registered elsewhere, eliminating duplicates and fraud that were previously undetectable for weeks.

2.8.2 Transformative Service Delivery

VERMS transforms service delivery from a slow, bureaucratic process into a efficient, citizen-centric experience.

1. **Instant Certificate Issuance:** The system eliminates weeks of waiting. Upon final approval, VERMS automatically generates a tamper-evident PDF certificate featuring a unique QR code, a digital signature of the issuing officer, and official seals. Citizens can download it instantly from the portal, receive it via email, or collect a printed copy from their local office, reducing process time from over a month to under an hour.
2. **Data-Driven Governance:** Supervisors and zone administrators are empowered with on-demand reporting and dashboard analytics. Instead of manual tallying that takes months, officials can generate accurate reports on birth rates, mortality, and marriage trends with a single click, enabling timely and evidence-based policy making and resource allocation.

2.8.3 Enhanced Security and Privacy Compliance

The proposed system directly addresses the critical security deficiencies of the current model.

- 1) **Robust Data Protection:** All sensitive citizen data will be encrypted in transit (TLS 1.2+) and at rest (AES-256) within a secure database, not in unlocked filing cabinets. Access is strictly controlled via Multi-Factor Authentication (MFA) and Role-Based Access Control (RBAC), ensuring staff only see data relevant to their duties.
- 2) **Full Accountability and Transparency:** An immutable audit trail logs every action from data entry to certificate generation creating a permanent record of who did what and when. This deters misuse and provides a clear chain of custody for legal purposes. Privacy is further enhanced by masking sensitive data in reports and providing citizens with transparency over who has accessed their records.

2.8.4 Preservation of Procedural Integrity

VERMS is designed to enhance, not replace, the foundation of the existing system.

1. **Business Rule Enforcement:** All existing business rules (BR-01 to BR-11) are hardcoded into the system's logic. Witness requirements, court order mandates for updates, and physical presence checks are transformed from manual procedures into automated, unforgettable system validations.
2. **Structured Workflow Integration:** The software mirrors and streamlines the existing hierarchical approval workflow (Kebele → Woreda → Zone), making the transition intuitive for existing staff. Their roles remain, but their tools are modernized, removing bottlenecks while preserving institutional knowledge.

2.8.5 A Scalable Platform for the Future

VERMS is not a static solution but a platform for continued innovation.

- **Future-Proof Architecture:** The system is built to allow for seamless integration of future enhancements, including:
 - a. SMS/Email Notifications for application status updates.
 - b. Mobile Registration Units with dedicated apps for reaching pastoralism communities.
 - c. API Integration's with national ID systems and healthcare databases for automated data verification and sharing.
 - d. Advanced Analytics for predictive modeling and advanced demographic insights.

By implementing VERMS, the South Gondar Zone will leapfrog from a struggling manual operation to a modern, efficient, and transparent digital service, setting a new standard for civil registration in Ethiopia. The subsequent chapters detail the technical specifications and implementation pathway to achieve this transformation.

CHAPTER 3: SOFTWARE REQUIREMENT SPECIFICATION

3.1 Introduction

Building upon the comprehensive analysis of the existing system's limitations in Chapter 2, this chapter formally defines the technical requirements and architectural design for the proposed Vital Events Registration Management System (VERMS). The specifications presented here translate operational needs into actionable technical blueprints while ensuring alignment with national digital governance standards.

3.2 General Constraints

The development and operation of VERMS are subject to the following constraints:

1. Technical Constraints:
 - ✓ The system must be a web application accessible on commonly available browsers (Chrome 70+, Firefox 65+).
 - ✓ It must be optimized to run on the existing hardware in Zone and Woreda offices (computers with minimum 4GB RAM).
2. Regulatory Constraints:
 - ✓ The system must fully comply with Ethiopia's Data Protection Proclamation (No.1205/2020) regarding the collection and storage of personal information.
 - ✓ The digital certificates generated must be considered legally admissible evidence in Ethiopian courts.
3. Operational Constraints:
 - ✓ The user interface must be bilingual (Amharic and English).
 - ✓ The system must be operable by staff with basic computer literacy.
 - ✓ Data migration from existing paper records is not in scope for the initial version.

3.3 Specific Requirements

3.3.1 External Interface Requirements

3.3.1.1 User Interfaces

1. Web-based Dashboard for Staff: A dashboard for South Gondar Zone Vital Events Registration Office staff (e.g., registrars, clerks, administrators) with features for data entry (e.g., registering vital events such as birth, death, marriage, divorce) and reporting (e.g., generating statistical and non-statistical reports). The interface must be intuitive, support Amharic and English languages, and be designed for users with minimal IT knowledge.

2. Citizen Portal: A web portal allowing citizens to request services (e.g., certificate issuance) and download certificates (e.g., birth, marriage certificates) after verification. The portal must be user-friendly and accessible on low-bandwidth networks.
- Accessibility Considerations: While the citizen portal is valuable for those with internet access, it is recognized that not all citizens have computers, smartphones, or literacy skills to use it. To address this, the following complementary access mechanisms will be provided:
1. Assisted Service at Registration Offices: Citizens without digital access can visit local registration offices, where clerks will assist them by using the staff dashboard to process their requests.
 2. Mobile Access (Optional Future Extension): For citizens with basic phones, services may be made accessible through USSD/SMS-based systems for simple requests (e.g., tracking application status).
 3. Help Desks & Community Support: Literacy barriers can be addressed by trained registrars and clerks, who guide citizens through the process when they come in person.
 4. Printed Awareness Materials: Posters and community campaigns in Amharic will help inform citizens about available digital and in-person service options.

3.3.1.2 Hardware Interfaces

- ✓ Compatibility: The system must be compatible with standard PCs and laptops used in South Gondar Zone offices (e.g., devices with specifications like 2.20 GHz CPU, 8GB RAM, 424GB hard disk).
- ✓ Support for Bar code Scanners: The system should interface with bar code scanners to facilitate document verification during registration or certificate issuance processes.

3.3.1.3 Software Interfaces

- ✓ Database Management: Integration with MySQL for efficient storage, retrieval, and management of vital event data, user accounts, and notifications, ensuring data integrity and scalability.
- ✓ Backend Technology: Use of PHP for server-side scripting to handle business logic, process requests, and manage database interactions securely.

- ✓ Frontend Technologies: Use of HTML, CSS, and JavaScript to build dynamic, responsive, and user-friendly interfaces for both staff and citizen portals, ensuring platform independence and ease of maintenance.

3.3.1.4 Communications Interfaces

- ✓ Secure Data Transmission: The system must use HTTPS to ensure encrypted communication between clients (e.g., web browsers) and the server, protecting sensitive data such as personal information and vital event records.
- ✓ APIs for Integration: The system should provide APIs to enable seamless integration with Kebele, Woreda, and Zone offices, supporting real-time data exchange and access to the centralized database.

3.4 Functional Requirements

Functional requirements are the intended features of the system. These features may be expressed as services, tasks, or functions that the system is required to perform. Functional requirement is a function or feature that must be included in an information system to satisfy the system's needs and be acceptable by the members. These are statements of service the system should provide, how the system should react to particular inputs, and how the system should behave in particular situations. It specifies the software functionality that the developer must build into the product to enable the user to accomplish tasks.

The following are functional requirements of our project:

System Administrator:

1. Create, update, and delete user accounts for all roles.
2. Assign and manage user roles and permissions.
3. Block and unblock user accounts when necessary.
4. Monitor system activities and generate audit logs.
5. Search account, creat backup

Kebele Registrar:

1. Register vital events (birth, death, marriage, divorce) at the Kebele level.
2. Update registered events when corrections are needed.
3. Search and View registered events.
4. Print and give certificates to citizens.
5. Receive feedback from zonal or woreda offices and respond or correct issues identified.

6. Generate Kebele-level reports for submission to Woreda offices.

Statistician:

1. Count event, view event and view report across Kebele and Woreda's within their zone.
2. Generate statistical reports (e.g., number of births, deaths, marriages, divorces).

Woreda Vital Events Registration Office Employees:

1. View reports, view registered events submitted by Kebele registrars.
2. Send feedback to Kebele registrars if problems are identified.
3. Receive feedback.
4. Generate woreda-level reports for submission to zonal offices.

Zonal Vital Events Registration Office Employees:

1. View registered event and view reports from multiple Woreda's and Kebele within the zone.
2. Send feedback to Kebele registrars through woreda offices regarding identified problems.
3. Generate zonal-level summary reports and forward them to the central statistics agency or regional office.

Citizen:

1. Request vital event registration services (birth, marriage, divorce, death) at the Kebele level.
2. Receive printed certificates and make payments for the requested services.
3. Provide feedback or complaints through the system interface.

3.5 Use Case Design

3.5.1 Actor Identifications

As we asked from South Gondar Zone Vital Events Registration Office employees and as we have seen from the organizational structure, the following are actors of the South Gondar Zone Vital Events Registration Information Management System:

1. System Administrator: A person who administers the system we are going to develop.
2. Kebele Registrar: A person responsible for registering vital events at the Kebele level.
3. Statistician: A person who works for the Zone's Central Statistics Agency, grouping population according to calculated information.

4. Wordage Vital Events Registration Office Employees: View the registered information and send feedback if any problems occur on forms.
5. Zonal Vital Events Registration Office Employees: View the registered information and send feedback if any problems occur on forms to the Kebele registrars.
6. Citizen: A person who requests vital event registration services (such as birth, marriage, divorce, and death) at the Kebele level and receives certificates or related documents.

3.5.2 Use Case Identification

Table 3.3: Use Case Identification

Use Case	ID	Precondition
Register events	UC01	Login
Update events	UC02	Login
View events	UC03	Login
Give/Receive certificate	UC04	Login
Generate report	UC05	Login
Count events	UC06	Login
Delete account	UC07	Login
Count birth	UC08	Login
Count death	UC09	Login
Count marriage	UC10	Login
Count divorce	UC11	Login
Send feedback	UC12	Login
Create account	UC13	Login
Block account	UC14	Login
Update account	UC15	Login
Search account	UC16	Login
View feedback	UC17	Login
View report	UC18	Login
Request event registration	UC19	Login
Assign roles/permissions	UC20	Login
Make payment	UC21	Login
Create backup	UC22	Login

Login	UC23	
Logout	UC24	

3.5.3 Use Case Diagram

A system use case model is composed of a use case diagram and the accompanying documentation describing the use cases, actors, and associations. The main difference between an essential use case and a system use case is that in the system use case, you include high-level implementation decisions.

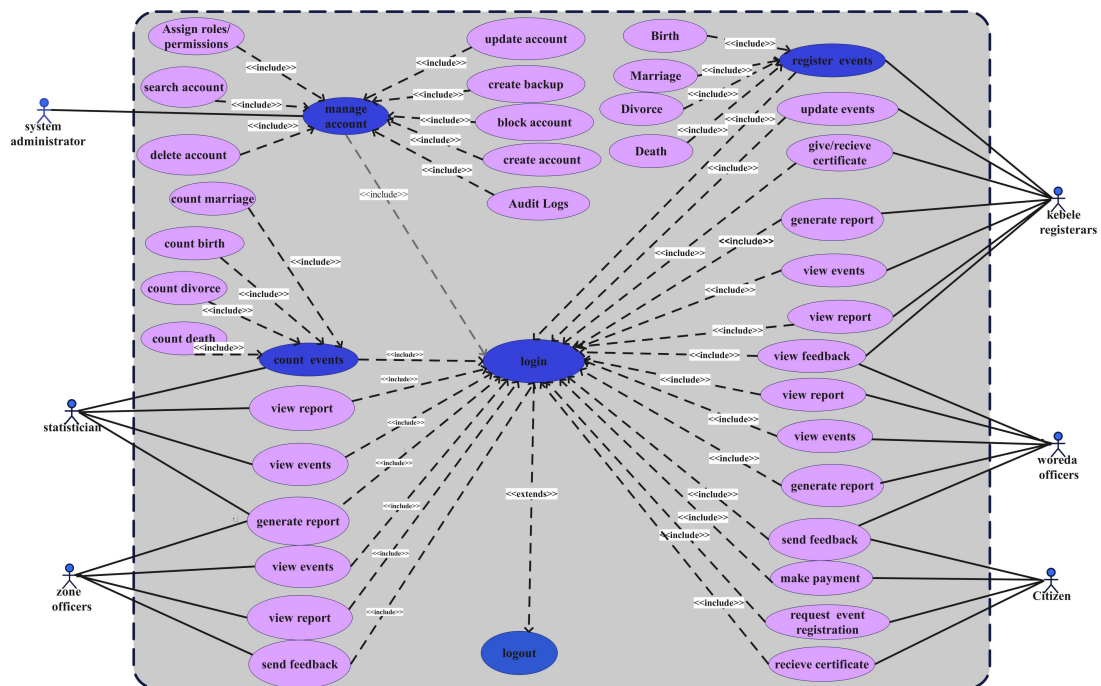


Figure 3.1: Use Case Diagram

3.5.4 Use Case Description

Table 3.4: Use Case Description and Actors

Actor	Use Case	Description
System Administrator	- Login - Create account - Block account - Update account - Search account -Assign roles/purmissions	Responsible for the technical management and maintenance of the VERMS. Manages user accounts, system configuration, and data backups. Has the highest level of system access.

	-delete account - Create backup -Audit Logs - Logout	
Kebele Registrar	- Login - Register events 1. Birth 2. Death 3. Marriage 4. Divorce - View events - Update events - Generate report - Give printed certificate - View feedback - View report - Logout	The primary data entry point located at the Kebele office. Responsible for initially registering all vital events with citizens. Give certificate to citizens.
Woreda Clerk	- Login - View events - Generate report - View report - Send feedback - Logout	Located at the Woreda (district) office. Responsible for validating, approving, and quality-checking records submitted by Kebele Registrars.
Zone Officer	- Login - View events - Generate report - View report - Send feedback - Logout	Located at the Zone head office. Responsible for final approval of complex cases, generating official certificates, and overseeing zone-wide operations.
Statistician	-Login -Count vital events 1. Count birth	A role often fulfilled by a Zone Officer or a dedicated analyst. Responsible for generating statistical reports and analyzing

	2. Count death 3. Count marriage 4. Count divorce -View events -View report -Generate report -Logout	demographic trends from the registered data.
Citizen	-Login -Request vital event registration -Receive certificate -Make payment -Send feedback -Logout	The end beneficiary of the services. Appears in person to register events or verify certificates. Interacts with the system indirectly through the registrars and the public portal.

Table 3.5: Use Case Description of Register Events

Field	Details
Name	Register event
Use case ID	UC01
Participating Actors	Kebele Registrar
Description	Used to register vital events at Kebele level.
Precondition	- The user must have an account to register events. - Have notification paper from health centers.
Post condition	- System adds event information to database. - Give certificate.
Basic course of action	1. The use case starts when registrars indicate they want to register events. 2. Then registrar enters username and password.

	3. Click login button. [Alt: A] 4. The system provides a list of operations. [BR11: access its own page only] 5. The Registrar selects register events link; it may be birth, death, marriage, divorce. 6. The Registrar fills and submits the form. [BR1, BR3, BR5, BR6, and BR7: register without error, do not duplicate events] [Alt: B] 7. The system receives data and prepares it to be printed on certificate if necessary. 8. The use case ends.
Alternative course of action	A: User entered invalid username and password A3: The system displays error message saying “Invalid user name or password.” A4: Use case continues from step 2 B: Registrar filled invalid form data B6: The system shows error message invalid form data message B7: System asks user to re-enter the invalid data again B8. Use case continues from step 6

Table 3.6: Use Case Description for Update Event

Field	Details
Name	Update events
Use case ID	UC02
Participating Actors	Kebele Registrar
Description	The registrars update the registered vital events.

Preconditions	- The user must bring order paper from courts that enables the registrar to update the registered event. - The registrar must have an account with update privilege.
Post conditions	The system successfully updates the vital events (birth, death, marriage, divorce).
Basic course of action	1. The use case starts. 2. Then registrar enters username and password. 3. Click login button. [Alt: A] 4. The system provides a list of operations. [BR11: access its own page only] 5. The Registrar selects update events link; it may be birth, death, marriage, divorce. 6. The registrar enters the event's identification number and submits. [Alt: B] 7. The Registrar updates and submits the form. [BR4, BR5, BR6: update events according to the order of courts and assure the correctness of data.] [Alt: C] 8. The system receives data and prepares it to be printed on certificate if necessary. 9. The use case ends.
Alternative course of action	A: User entered invalid username and password A3: The system displays error message saying "Invalid user name or password." A4: Use case continues from step 2 B: Registrar filled invalid form data B6: The system shows error message "invalid ID number". B7: System asks user to re-enter the invalid

	data again B8. Use case continues from step 6 C: Registrar fills the data that is needed to be updated. C7: The system displays error message saying “Invalid form data.” C8: System asks user to re-enter the invalid data again. C9: Use case continues from step 7
--	---

Table 3.7: Use Case Description for give Printed Certificate

Field	Details
Use case name	Give printed certificate
Use case ID	UC04
Participating Actors	Kebele Registrar
Description	The registrars give printed certificates by the request of the customers.
Preconditions	The event must exist in the database, and the certificate is printed if the end user requests.
Basic course of action	1. The use case starts. 2. The actor enters username and password. 3. Click login button. [Alt: A] 4. The system provides a list of operations. [BR11: access its own page only] 5. The actor selects generate certificate link. 6. Retrieve necessary information from the database. [Alt: B] 7. Click on print button. 8. The use case ends.
Alternative course of action	A: User entered invalid username and password A2: The system displays error message

	saying “Invalid user name or password.” A3: Use case continues from step 2 B: Registrar retrieves invalid data. B6: The system shows error message “information does not exist”. B7: System asks user to re-enter the valid data again B8. Use case continues from step 6
Post conditions	Give printed certificate filling correct information to end users.

Table 3.8: Use Case Description for Generate Report

Field	Details
Use case name	Generate report
Use case ID	UC05
Participating Actors	-Statistician, Kebele Registrar, Zone Officer , Woreda Officer
Description	- Generating reports that give general information about the records in the database.
Preconditions	The actor must have an account and privilege to generate reports.
Basic course of action	1. The registrar enters username and password. 2. The registrar clicks the login button. [Alt: A] 3. The system provides a list of operations. [BR11: access its own page only] 4. The registrar selects generate report link. 5. The registrar selects the report type, then generates report. [Alt: B]

	6. The use case ends.
Alternative course of action	A: User entered invalid username and password A2: The system displays error message saying "Invalid user name or password." A3: Use case continues from step 2 B: Registrar wants to retrieve data that is not in the database. B6: The system shows error message "information does not exist". B7: System asks user to retrieve again. B8. Use case continues from step 5
Post conditions	Generate reports successfully.

Table 3.9: Use Case Description for Login

Field	Details
Use case name	Login
Use case ID	UC23
Participating Actors	Citizen, Kebele Registrars, Statistician, Zone Officers, Woreda Officers, Administrator
Description	Users are authenticated and enter their own user interface page.
Precondition	Users must get an account from the System Administrator.
Post condition	The user is authenticated and taken to their own user interface.
Alternate course of action	A: Invalid Username or Password A1: If the username or password is invalid, the system displays an "Invalid username or password" message. A2: The user re-enters the username and

	password. A3: The system returns to step 4 of the main flow. B: Account Locked B1: If the user exceeds the maximum number of login attempts (e.g., 3), the system locks the account and displays a "Account locked, contact administrator" message. B2: The user contacts the System Administrator to unlock the account. C: System Unavailable C1: If the system is down or unreachable, a "System unavailable, try again later" message is displayed. C2: The user retries later.
--	---

3.6 Sequence Diagram

A sequence diagram shows the sequence of interactions among objects and is used to represent or model the flow of messages, events, and actions between the objects or components of a system. Sequence diagrams are also used primarily to design, document, and validate the architecture and interfaces of the system by describing the sequence of actions that need to be performed to complete a task.

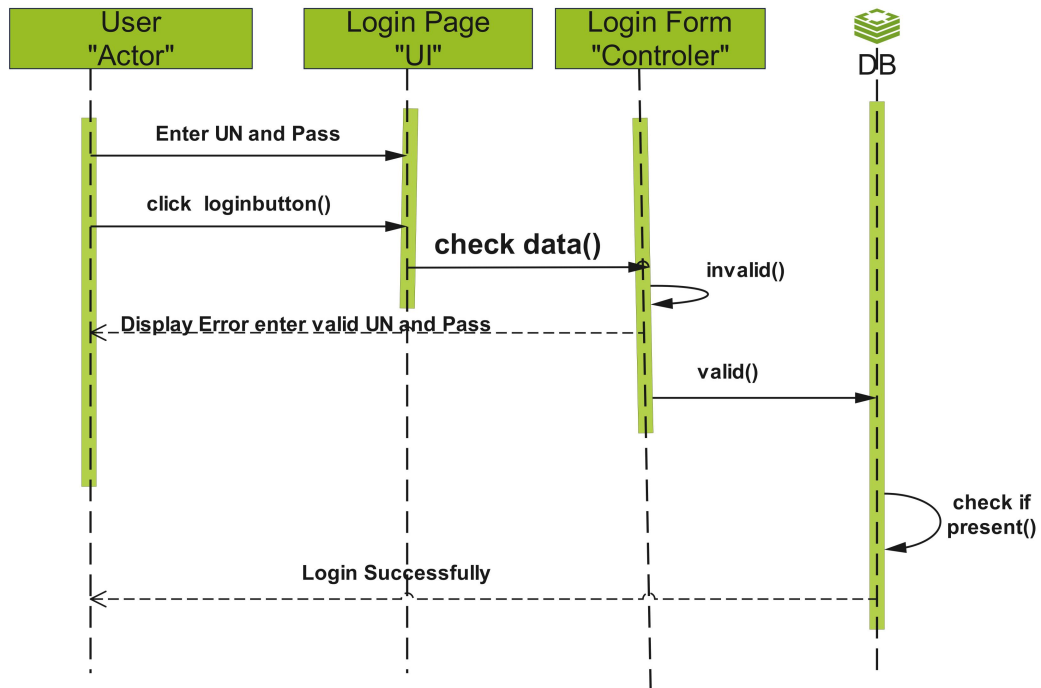


Figure 3.2: Sequence Diagram for Login

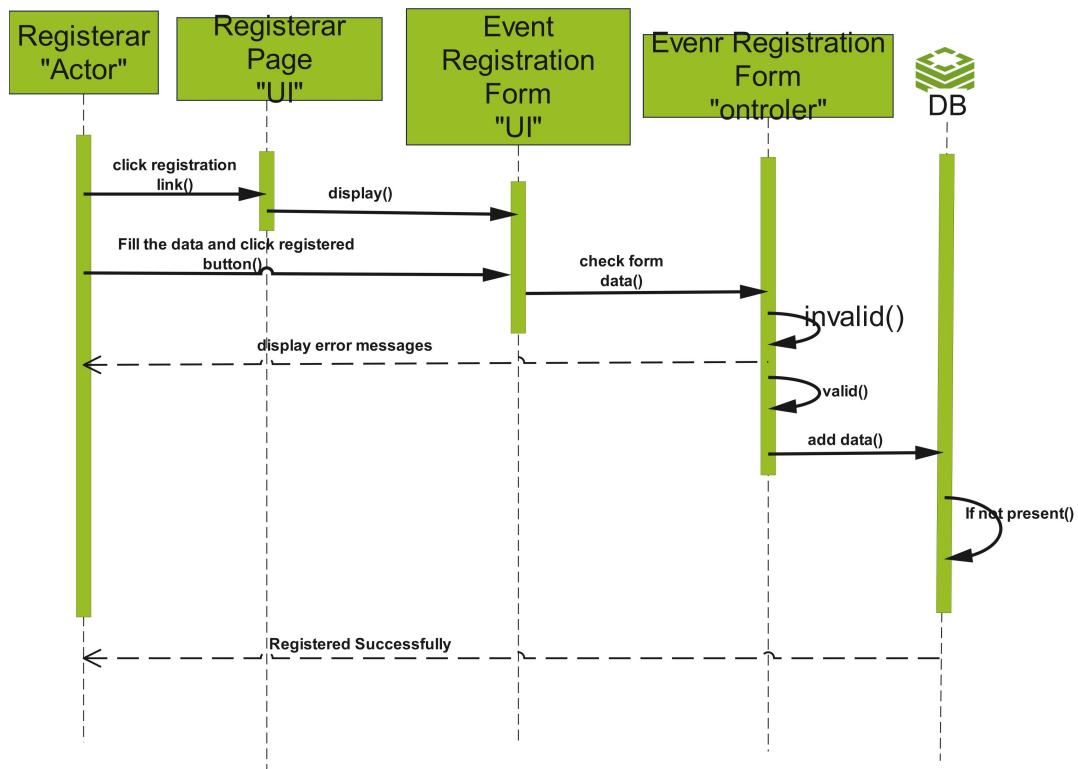


Figure 3.3: Sequence Diagram for Registering Events

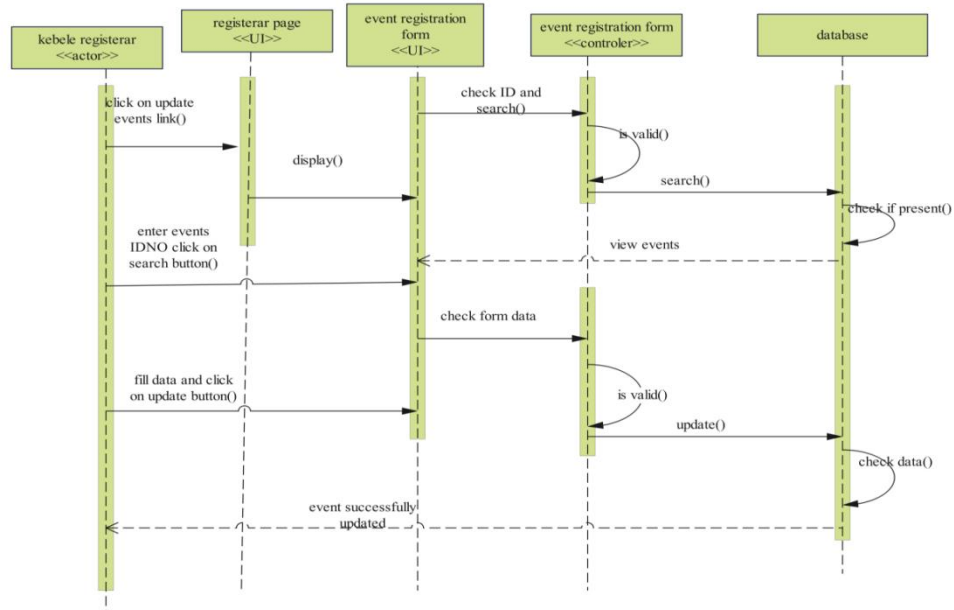


Figure 3.4: Sequence Diagram for Updating Events

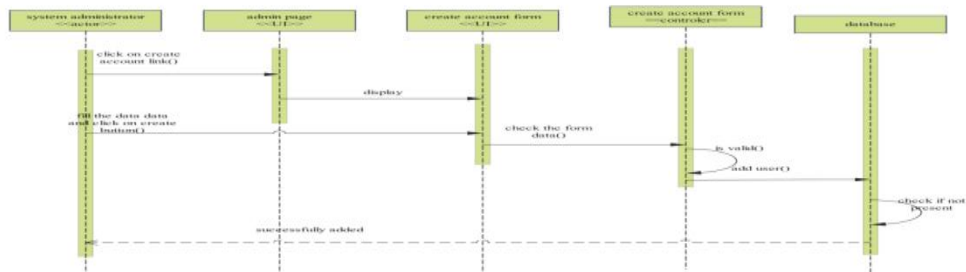


Figure 3.5: Sequence Diagram for Create User Account

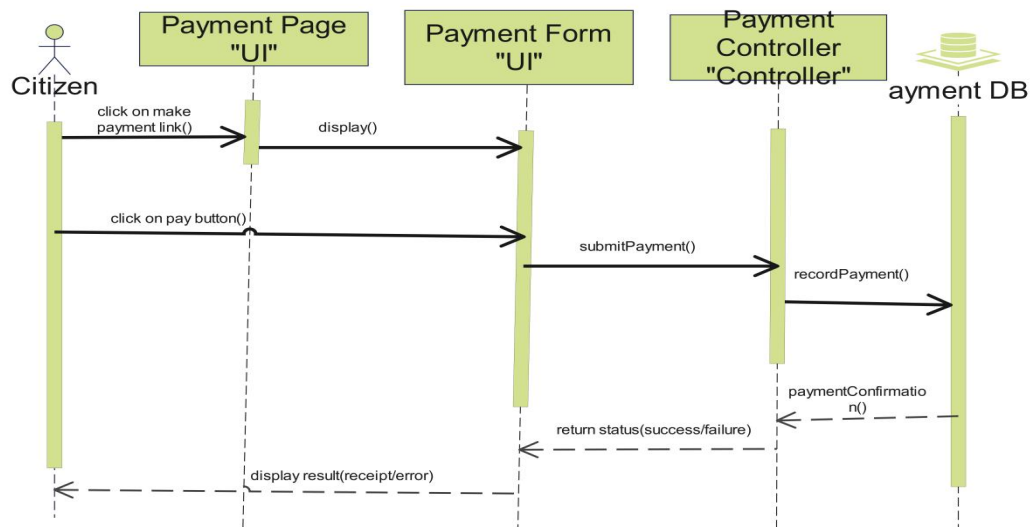


Figure 3.6: Sequence Diagram for Make Payment

3.7 Activity Diagram

An activity diagram is basically a flowchart to represent the flow from one activity to another. The activity can be described as an operation of the system. So the control flow is drawn from one operation to another. This flow can be sequential, branched, or concurrent. Activity diagrams deal with all types of flow control by using different elements like fork, join, etc.

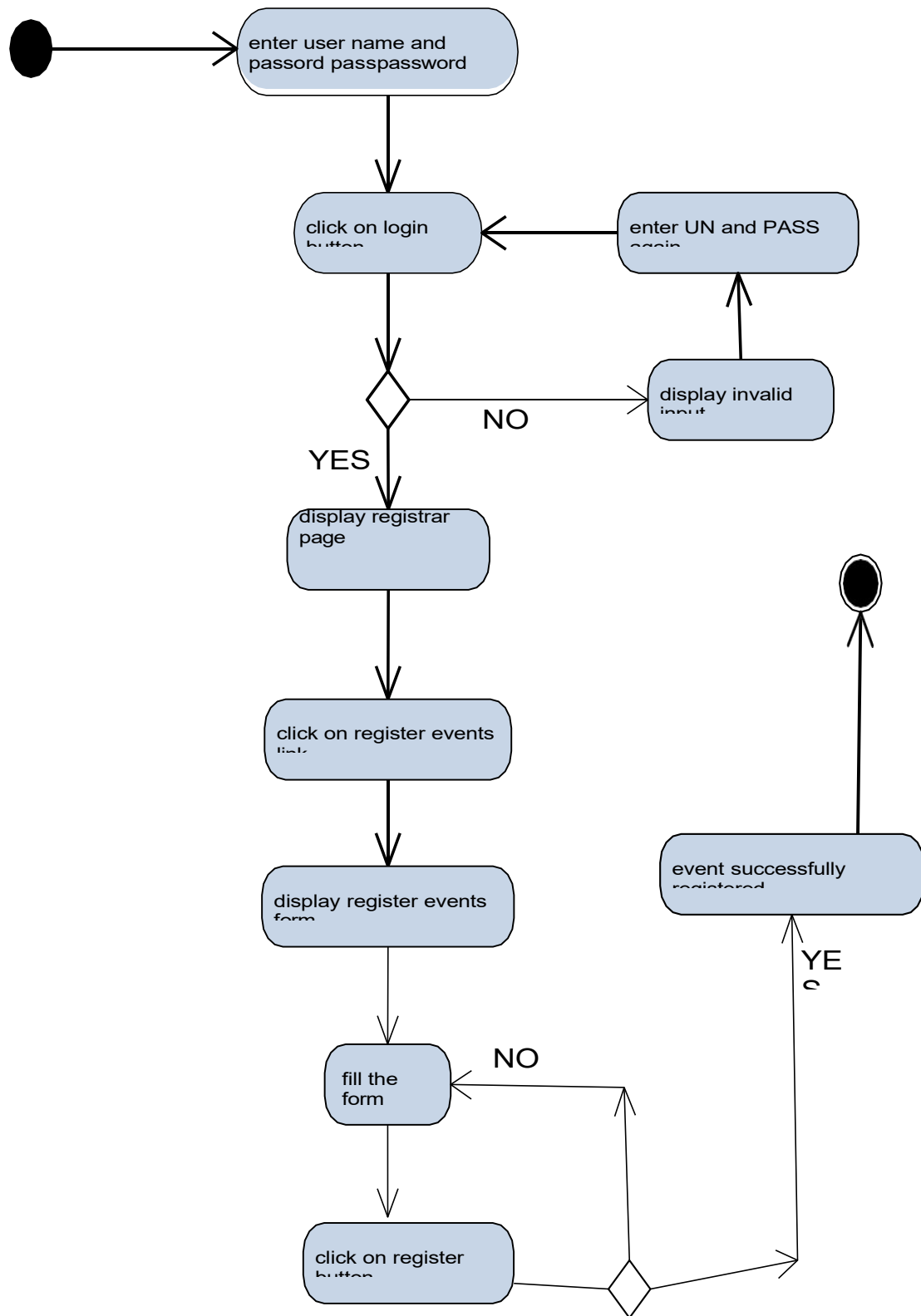


Figure 3.9: Activity Diagram for Register Events

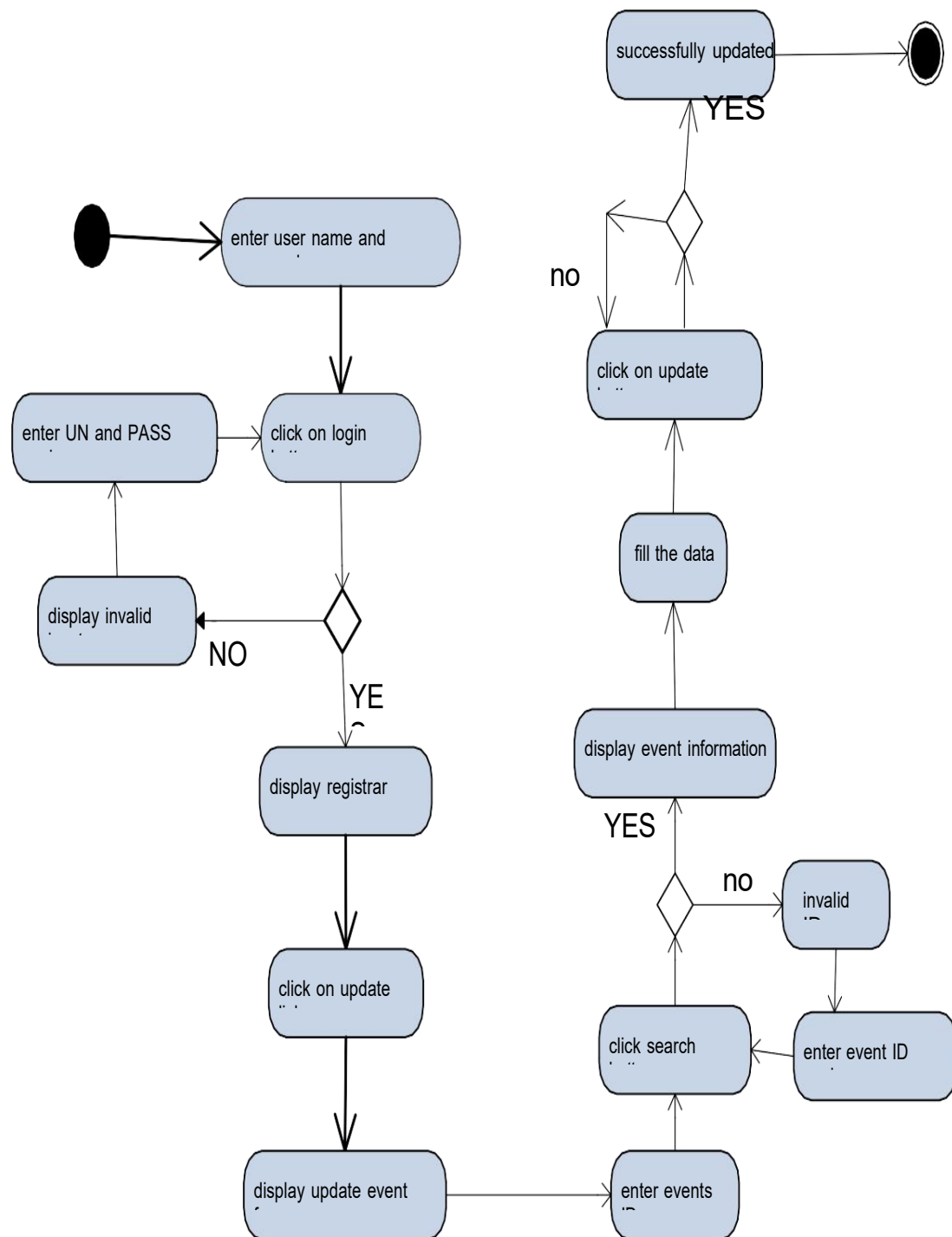


Figure 3.10: Activity Diagram For Update Event

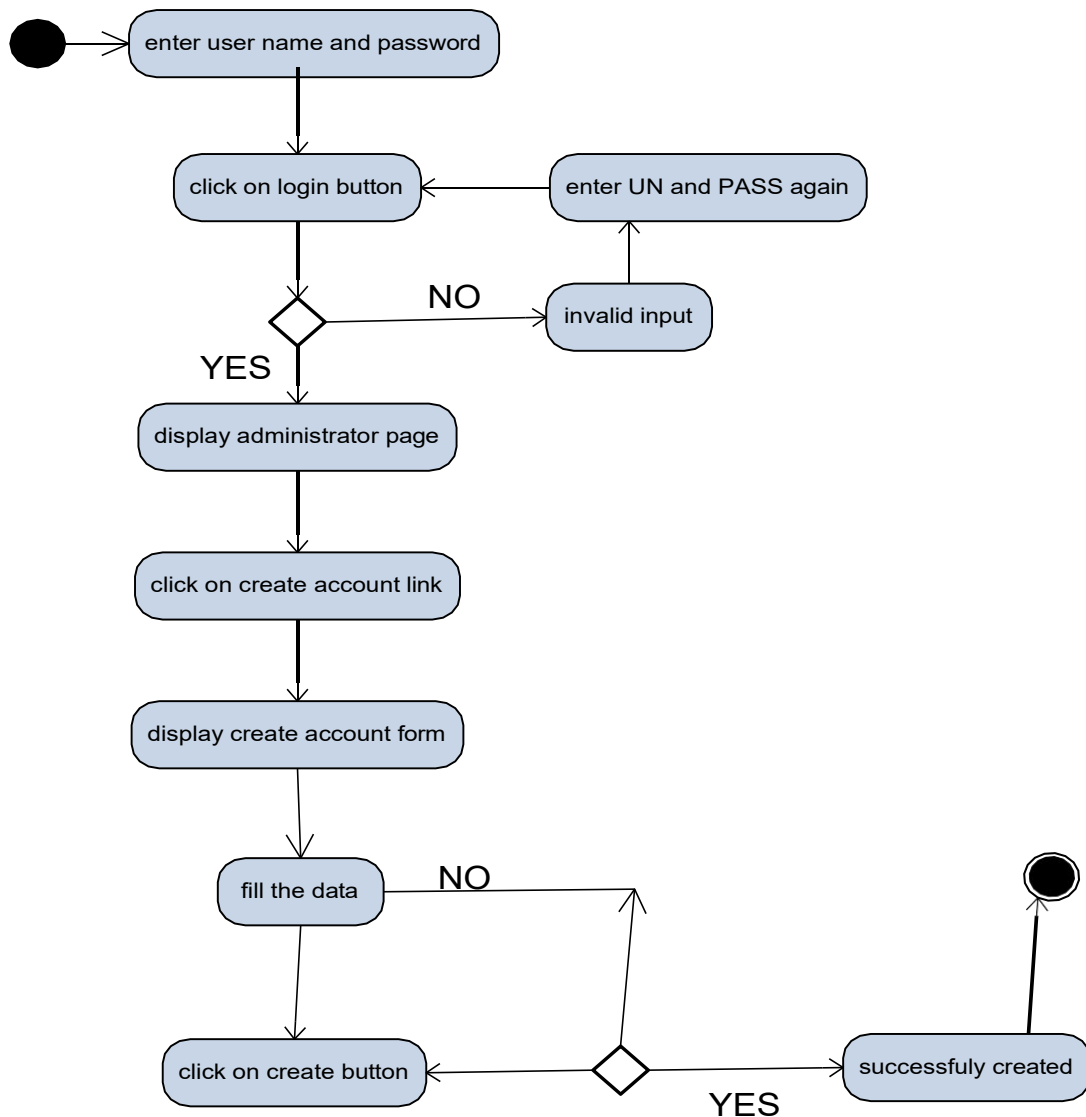


Figure 3.11: Activity Diagram for Create Account

3.8 Class Diagram

Class diagrams are the mainstay of object-oriented modeling. Class models show the classes of the system, their interrelationships (including inheritance, aggregation, and association), and the operations and attributes of the classes. Class diagrams are used for a wide variety of purposes, including both conceptual/domain modeling and detailed structural design modeling.

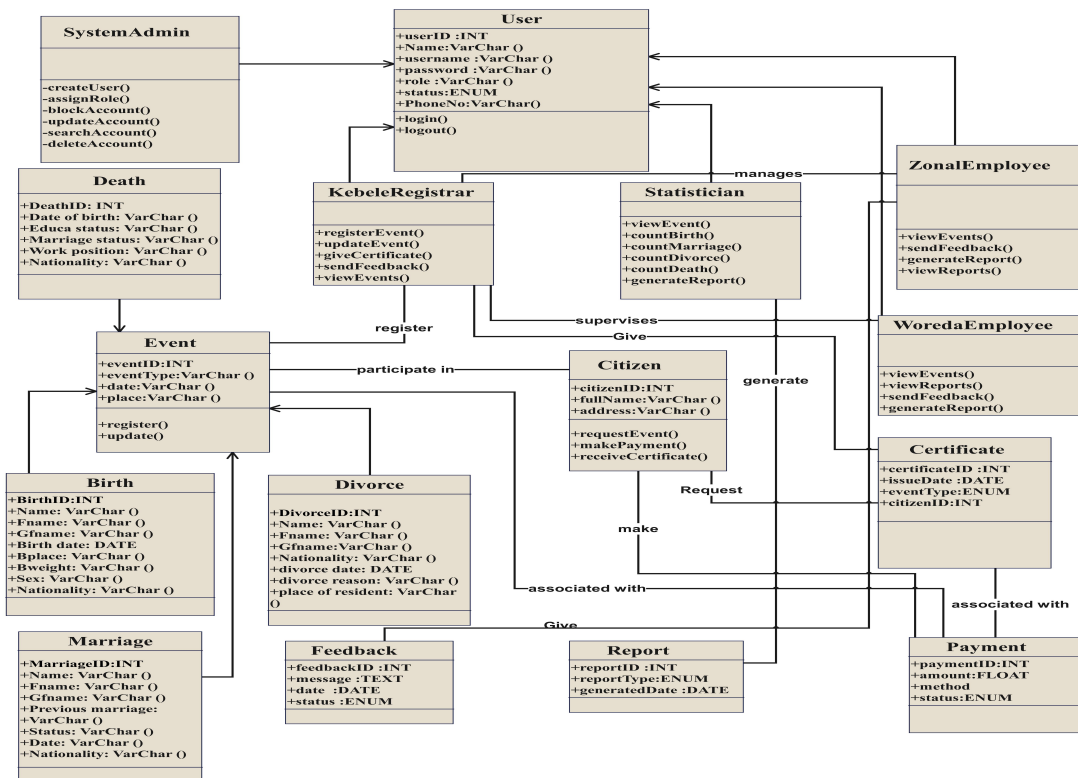


Figure 3.12: Class Diagram

3.9 Non-Functional Requirements

Non-functional requirements specify criteria that can be used to measure the quality of a system, rather than specific behaviors. This should be compatible with functional requirements that define specific behavior or functions.

In general, functional requirements define what a system is supposed to do, whereas non-functional requirements define how a system is supposed to be. Non-functional requirements are often called qualities of a system. Other terms for non-functional requirements are constraints, quality attributes, quality goals, and quality of service requirements.

Qualities that are non-functional requirements are the following:

- Performance: Performance is characterized by the amount of tasks that are accomplished by a system within a given time and resources used.

Good performance may involve one or more of the following characteristics:

1. Short response time for a given task.
2. High efficiency.
3. High availability means that the system must stay up 24 hours a day within 7 days in a week.
4. High data transmission rate.

To improve the performance of the system that we are going to develop, we will use the following methods:

- 1) Installing our system on multiple servers.
- 2) We will use high-capacity processors.
- 3) Improving bandwidth to make short response times.

➤ Security:

To make the system safe from unauthorized users, the system will use a login account to differentiate authorized users from unauthorized users of the system. This enables the system to verify who has logged in.

1. We will use two-layer authentication (username, password).
2. We will restrict the privilege of users; for example, Woreda Vital Events Registration Office employees must access only their Woreda's data.
3. We will use sessions to restrict users from accessing pages without their privilege, so we will give session times that will expire after the time passes.
4. We will protect the system from external intruders by using password encryption methods like MD5, HASH, SHA1, etc.

➤ Usability:

Usability refers to how a system is easy or clear to interact with users. The system should be easy to learn and operate with the existing staff.

It also includes the following points:

- a. More efficient to use it takes less time to accomplish a particular task.
- b. Easier to learn operation can be learned by observing the object.
- c. More satisfying to use.

To improve our system's usability, we will introduce the following methods:

- a) Designing a more interactive user interface by using more efficient codes like the latest versions of CSS and HTML.
- b) We will make the system more usable by employees of the Vital Events Registration Office by providing continuous training.
- c) We will develop a system that will work with at least two languages: Amharic and English.

➤ Reliability:

The system is expected to perform perfectly without major errors and failures. The system should satisfy the user and be fault-tolerant.

To achieve this:

- a. We will install our system on high-quality hardware.
- b. Back up regularly.
- c. Control external factors like power loss as much as possible.

➤ Modifiability:

The system that we will develop will be easy to modify by its users if necessary.

- a) If we want to add new functionalities.
- b) If we want to add new features, the system will continue performing as it did previously, even if we change some features.

CHAPTER 4: SYSTEM DESIGN

4.1 Design Overview

The design of the South Gondar Zone Vital Events Registration Office Information Management System aims to transform the manual, paper-based vital events registration process into an automated, web-based system. The system addresses the registration, updating, viewing, and reporting of vital events (birth, death, marriage, divorce) across Kebele, Woreda, and Zone levels. The primary goal is to eliminate inefficiencies, reduce errors, ensure data integrity, and provide a centralized database accessible to authorized users at these levels.

- Key design objectives include:
 - ✓ Automation: Replace paper-based processes with digital forms to streamline registration and data management.
 - ✓ Accessibility: Provide role-based access to a centralized database for Kebele registrars, statisticians, and officers at Woreda and Zone levels.
 - ✓ Data Integrity: Enforce business rules (e.g., BR1: no duplicate events, BR3: error-free registration) through form validations and constraints.
 - ✓ Usability: Develop an intuitive user interface using HTML, CSS, and JavaScript to ensure ease of use for users with minimal IT knowledge.
 - ✓ Scalability and Security: Implement a client-server architecture with MySQL and PHP, ensuring secure data transmission and scalability for multiple users.

The system design integrates functional requirements (e.g., event registration, certificate generation) and non-functional requirements (e.g., performance, security, usability) to meet the needs of the South Gondar Zone Vital Events Registration Office. The design process involves defining the system architecture, user interfaces, and database structure.

4.2 Chosen System Architecture

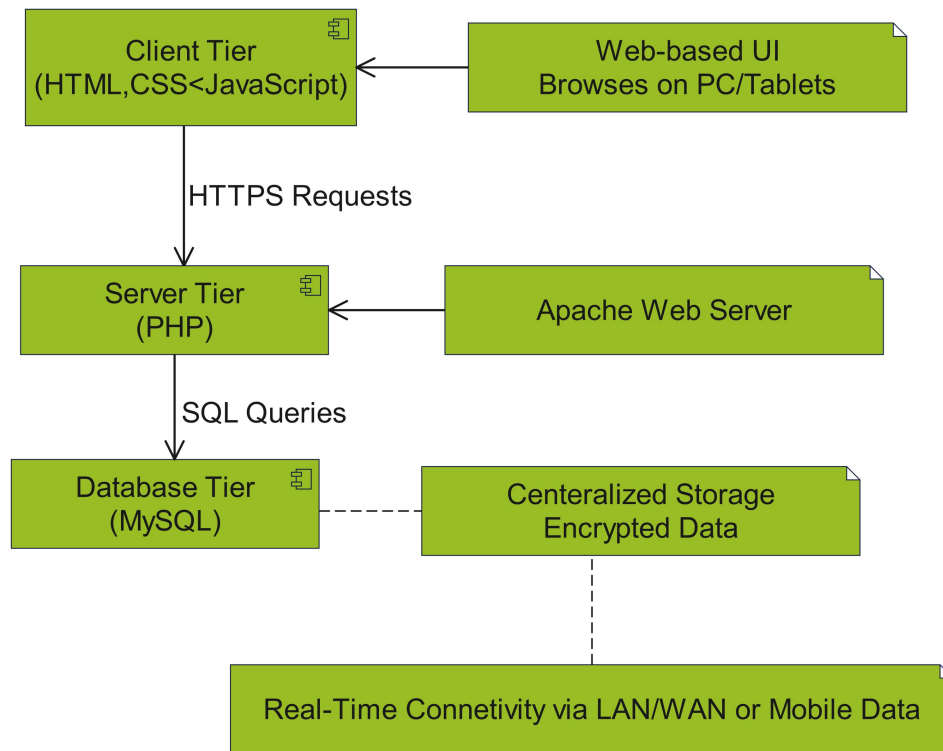


Figure 4.13: Detailed System Architecture Diagram

➤ Architecture Components:

✧ Client Tier:

- Consists of web browsers (e.g., Mozilla Firefox) running on client devices (e.g., PCs with 2.20 GHz CPU, 8GB RAM, 424GB hard disk).
- Clients access the system through user interfaces tailored to specific roles (e.g., system admin page, Kebele registrar page, statistician page).
- Front-end technologies include HTML, CSS, and JavaScript, ensuring a user-friendly and platform-independent interface.

✧ Server Tier:

- Hosted on a dedicated server running a WAMP stack (Windows, Apache, MySQL, PHP).
- The server handles business logic, data processing, and database interactions using PHP as the server-side scripting language.
- MySQL serves as the relational database management system, storing vital event data, user accounts, and notifications.

➤ **Network Infrastructure:**

The system requires a network to connect offices at Kebele, Woreda, and Zone levels, enabling real-time data transmission and access to the centralized database.

Security measures include two-layer authentication (username and password), session management, and password encryption (e.g., MD5, SHA1) to protect against unauthorized access.

➤ **Rationale for Selection:**

- ✓ **Scalability:** The client-server model supports multiple concurrent users across the South Gondar Zone's administrative levels.
- ✓ **Centralized Data Management:** A single MySQL database ensures data consistency and eliminates the need for paper-based data transfers between offices.
- ✓ **Ease of Maintenance:** PHP and MySQL are open-source, widely supported, and easy to maintain, aligning with the project's economic feasibility.
- ✓ **Role-Based Access:** The architecture supports business rule BR11 (employees access the system based on their privilege) through role-specific pages and session restrictions.
- ✓ **Deployment Flexibility:** The deployment diagram illustrates how software artifacts are distributed across client pages, ensuring flexibility for different user roles.

4.3. User Interface Design

4.3.1. Description of the User Interface

The user interface (UI) is designed to be intuitive, interactive, and accessible to users with varying levels of technical expertise, including Kebele registrars, zone officers, statisticians, woreda officers and administrative officers. The UI is built using HTML, CSS, and JavaScript, ensuring a responsive and visually appealing experience.

4.3.2. Screenshot Images

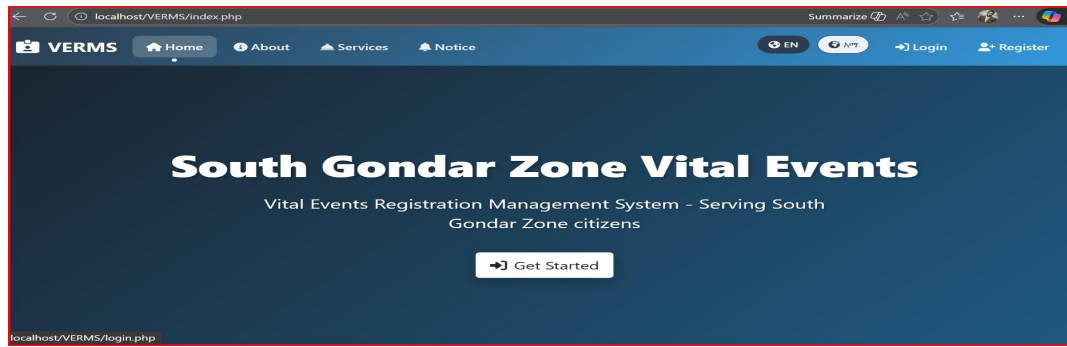


Figure 4.14 Home Page

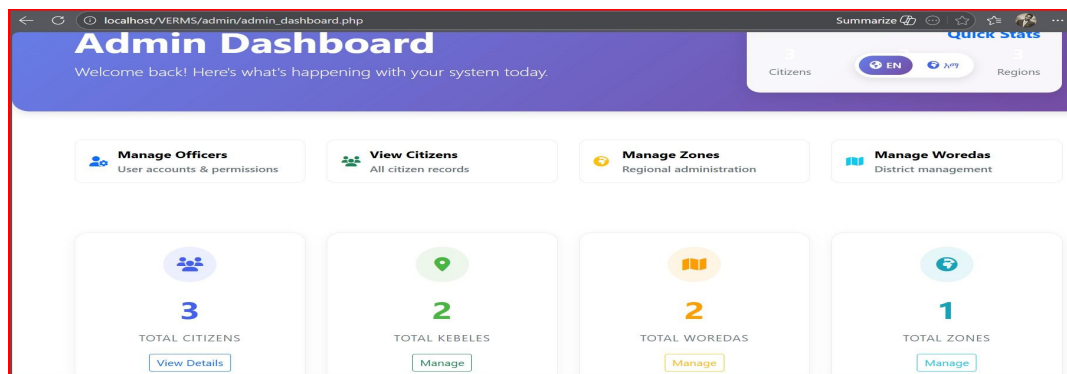


Figure 4.15 Admin Page

4.4 Database Design

A Persistent Diagram represents the structure of data as it is intended to be maintained across application sessions within a specific database management system. Within the project life cycle, it acts as the bridge between the Logical Data Model and the actual Data Access Layer, defining how domain objects are mapped to permanent storage [6].

A comprehensive persistent diagram encompasses all artifacts necessary for data integrity and retrieval efficiency, including:

1. Entity Mappings: Transitioning from logical entities to physical table structures.
2. Attribute Persistence: Specific column names, data types, and nullability constraints.
3. Identity Management: Implementation of Primary Keys (e.g., UUIDs vs. Auto-increments) and Foreign Key constraints to enforce referential integrity.
4. Optimization Artifacts: Definitions for indexes, partitioned tables, and clusters designed to meet specific performance benchmarks.
5. State Management: Details regarding storage allocation and partitioning strategies used by analysts to calculate data volume estimates

4.4.1. Persistent (ER) Diagram

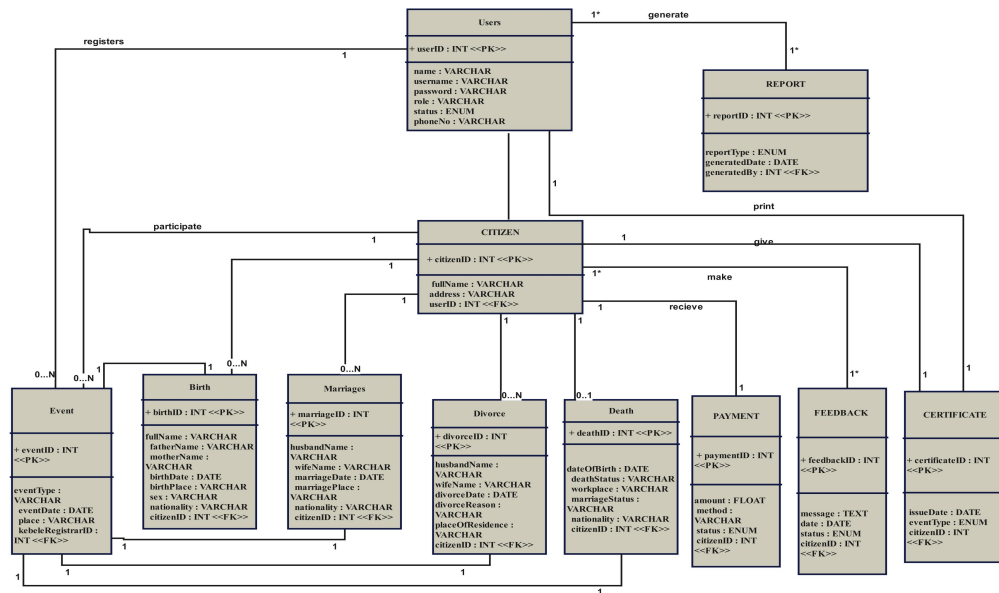


Figure 4.16: Persistent ER Diagram

4.4.2. Normalization

Normalization is the process of organizing data in a database to minimize redundancy, eliminate anomalies (such as insertion, update, and deletion issues), and ensure data integrity. It involves dividing large tables into smaller, related tables and defining relationships between them using primary and foreign keys. The goal is to achieve higher normal forms, typically up to the Third Normal Form (3NF) for most practical databases, where:

- First Normal Form (1NF): Ensures atomic (indivisible) values in each column, no repeating groups, and each row is unique.
- Second Normal Form (2NF): Builds on 1NF by ensuring no partial dependencies (all non-key attributes depend on the entire primary key).
- Third Normal Form (3NF): Builds on 2NF by ensuring no transitive dependencies (non-key attributes do not depend on other non-key attributes).

The database design for VERMS, as outlined in the persistent diagram is already structured to comply with these normal forms.

CHAPTER 5: TESTING

5.1 Test Plan

5.1.1 Responsibilities of each Group Member

Table 5.10: Responsibilities of each Group Member

Role	Team Member	Key Responsibilities
Test Manager	Tesfaw	Develops the master test plan, manages the testing schedule, allocates resources, and communicates overall progress.
Test Lead / QA Engineer	Helen	Writes detailed test cases, executes system and usability tests, logs defects, and generates test summary reports.
Security & Performance Tester	Fasika	Conducts security vulnerability scans (OWASP ZAP), performance tests, and validates offline functionality.
Database Tester	Mekdes	Verifies data integrity, tests SQL queries, ensures proper audit logging, and validates backup/restore procedures.
Development Tester	Dawit	Develops and executes unit and integration tests; fixes bugs identified by the QA team.
UAT Coordinator	Helen	Trains and supports Zone office staff during User Acceptance Testing (UAT) and collects their feedback.

5.1.2 Types of Testing

The following testing types will be performed to ensure comprehensive coverage:

1. **Unit Testing:** Testing individual components or functions in isolation (e.g., testing a function that calculates age from date of birth).
2. **Integration Testing:** Testing the interaction between integrated modules (e.g., ensuring the frontend form correctly sends data to the backend API, which then saves it to the database).
3. **System Testing:** Testing the complete, integrated system to verify that it meets the specified requirements (Black-box testing).

4. **User Acceptance Testing (UAT):** Testing conducted by the end-users (Zone staff) to ensure the system meets their business needs and is ready for operation.
5. **Performance Testing:** Testing the system's responsiveness and stability under expected load (e.g., 50 concurrent users).
6. **Security Testing:** Testing for vulnerabilities like SQL Injection, Cross-Site Scripting (XSS), and ensuring proper access controls.
7. **Usability Testing:** Testing the user interface for intuitiveness, ease of use, and compliance with Amharic and English language support.
8. **Compatibility Testing:** Testing the application on different browsers (Chrome, Firefox) and devices.
9. **Regression Testing:** Re-testing after fixes or changes to ensure no existing functionality was broken.

5.1.3 Features to be Tested

All features derived from the Functional Requirements (Chapter 3) will be tested:

1. Vital Event Registration (Birth, Death, Marriage, Divorce)
2. Record Management and Audit Trail
3. Certificate Generation with QR Code
4. Reporting and Data Analytics
5. Public Verification Portal
6. Cross-browser compatibility on Chrome and Firefox.

5.1.4 Features not to be Tested

1. **Payment Processing Integration:** This is a future feature and will be simulated, not fully implemented, in this release.
2. **Integration with external national systems:** (e.g., National ID database, Health Ministry systems). These are out of scope.
3. **Load testing beyond 75 concurrent users.**

5.1.5 Testing Tools

Table 5.11: Testing Tools

Testing Type	Tool(s)	Purpose
Unit Testing	PHP Unit , Jest	To test individual functions and classes.
Integration/API	Postman, HTTP Tests	To test API endpoints and data

Testing		flow between modules.
Browser Testing	Selenium	To automate and run UI tests in browsers.
Performance Testing	Apache JMeter	To simulate multiple users and measure system performance.
Security Testing	OWASP ZAP	To identify security vulnerabilities like SQL injection.
Usability/Accessibility	Lighthouse, manual testing	To check for accessibility issues and overall user experience.
Defect Tracking	GitHub Issues	To log, track, and manage bugs throughout the testing life cycle

5.2 Test Cases

Scenario id: Tc-auth-01

Test case description: Verify login fails with invalid password and provides appropriate error message.

Test priority: High

Action: 1. Navigate to login page. 2. Enter valid username. 3. Enter invalid password. 4. Click 'log in'.

Inputs: Username = kebele1, password = Kebele@12

Expected output: System displays a generic error message: "Invalid password." The system does not reveal whether the username or password was incorrect. User is not logged in.

Actual
output:

The image displays two screenshots of the VERMS (Vital Events Registration Management System) login interface. The top screenshot shows a user successfully logging in with the username 'kebele1', password 'Kebele@1212', and role 'Kebele Officer'. The bottom screenshot shows an 'Invalid password' error message displayed in a red box, indicating a failed login attempt.

Scenario id: Tc-regist-02

Test case description: Verify successful registration of a death event with all required fields.

Test priority: High

Action: 1. Log in as Kebele registrar. 2. Select 'death registration'. 3. Complete all mandatory fields. 4. Submit the form.

Inputs: Deceased full name = Wro. Alemitu fisseha, date of death = 02/12/2024, cause of death = illness, informant name = teklu alemayehu, Relationship to Deceased= family

Expected output: System displays success message: "Death registration successful.The new record appears in the 'my registrations' list with the 'pending' status.

Actual output:

Recent Events			
Event	Name	Status	Date
DEATH	Wro. Alemitu fisseha	Pending	Jan 11, 2026

Scenario id: Tc-regist-03

Test case description: Verify form validation prevents submission with missing required fields.

Test priority: High

Action: 1. Log in as Kebele registrar. 2. Select 'birth registration'. 3. Enter only the child's first name. 4. Click 'submit'.

Inputs: Child first name = solomon

Expected output: The form is not submitted. System highlights the empty mandatory fields (e.g., last name, date of birth, father's name) with red borders and displays error messages: "This field is required."

Actual

output:

The screenshot shows a web browser window with the URL `localhost/VERMS/kebele_officer/birth.php`. The form is titled "Child Information" and contains several sections:

- Child Information:** Includes fields for "Child Full Name" (filled with "solomon"), "Father Name" (empty), and "Grandfather" (empty). The "Date of Birth" field is filled with "01/11/2026" and has a red border and an error message "Please fill out this field." above it.
- Parents Information:** Includes fields for "Mother Full Name" and "Father Full Name", both of which are empty.
- Registration Details:** Includes fields for "Birth Registered Date" (filled with "01/11/2026") and "Certificate Issue Date" (filled with "01/11/2026").

The form also includes a "Sex" dropdown menu set to "Select" and a "Parents' Nationality" dropdown menu set to "Ethiopian".

Scenario id: Tc-access-04

Test case description: Verify kebele registrar cannot access zone officer's dashboard page (role-based access control).

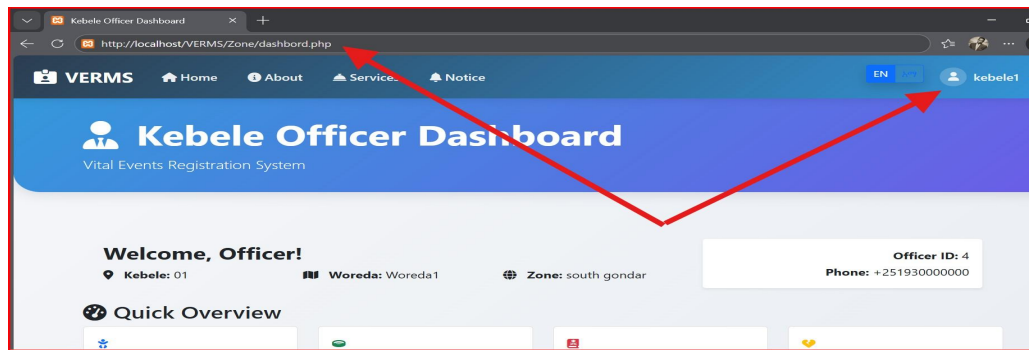
Test priority: High

Action: 1. Log in with Kebele registrar credentials. 2. Manually type the URL for the dashboard page (`/Zone/dashboard`).

Inputs: URL = `http://localhost/VERMS/Zone/dashboard.php`

Expected output: The user remains on their dashboard or is redirected to its own a safe page.

Actual Output:



Scenario id: Tc-cert-05

Test case description: Verify Print pdf certificate contains correct data and a scannable qr code.

Test priority: Medium

Action: 1. Log in as Citizen. 2. Find an approved birth record in view Certificate. 3. Click Print certificate'. 4. Open the PDF and scan the qr code.

Inputs: Registration number = FORM-mk87rnph-UR4US

Expected output: A pdf file downloads automatically. The PDF contains the correct child's name, date of birth, registration number, and official seals. Along with public details of the record.

Actual Output:



Scenario id: Tc-report-06

Test case description: Verify generation of a birth report.

Test priority: Medium

Action: 1. Log in as Kebele officer. 2. Navigate to reports and analysis. 3. Select 'birth report'. 4. Click 'generate report'. 5. Select 'export as pdf'.

Inputs: Period = January 13, 2026.

Expected output: A PDF report is generated. The report title is "Birth report for Kebele01. The report contains a table listing all birth registrations from

Kebele01 in that period, including child name, date of birth, and Kebele.

Actual output:

BIRTH EVENTS REGISTER						
01 Kebele						
Report Period: January 13, 2026						
#	Child Name	Father Name	Mother Name	Date of Birth	Sex	Registered Date
1	ከላለድህድ	ሰድፍ ድጅሁዊ	አረመኑ ጅህረኢ	Jan 10, 2026	Female	Jan 10, 2026
2	alemu	Tamiru	Betelhem	Jan 6, 2026	Male	Jan 10, 2026

Generated by VERMS on January 13, 2026 at 7:56 AM

Scenario id: Tc-usab-07

Test case description: Verify UI language can be switched to Amharic.

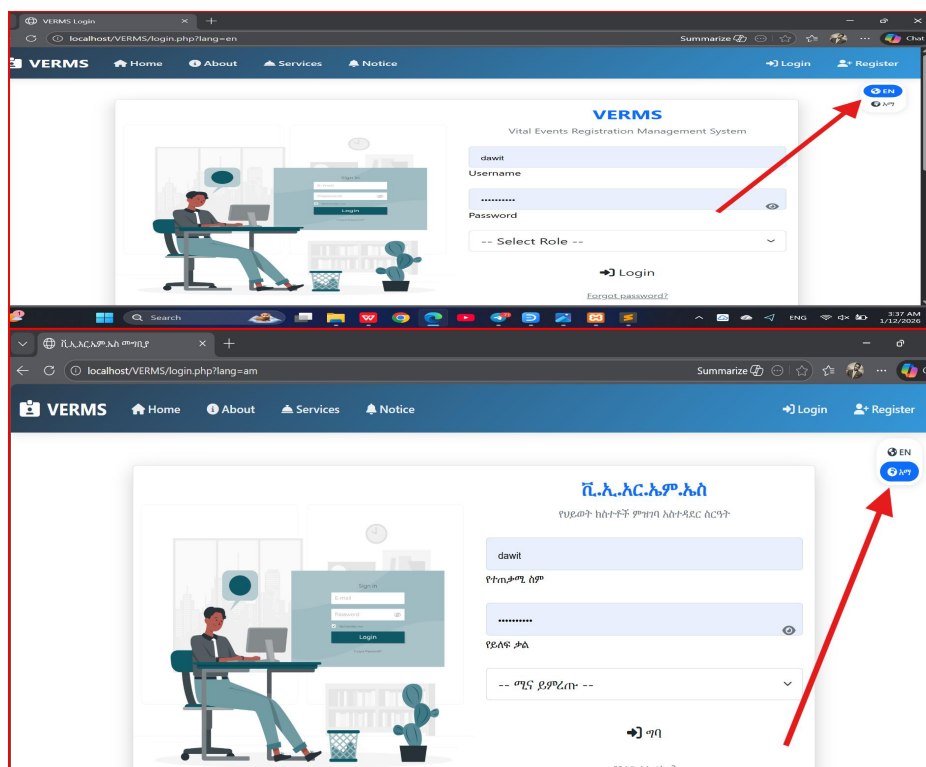
Test priority: Medium

Action: 1. Navigate to the login page. 2. Click the language selector dropdown. 3. Select "አማርኛ (amharic)".

Inputs: N/a

Expected output: The entire user interface, including labels, buttons, and messages, instantly changes to Amharic. The page title and content are correctly displayed in Amharic script.

Actual output:



CHAPTER 6: CONCLUSION AND RECOMMENDATION

6.1 Conclusion

This project successfully undertook the comprehensive analysis, design, and planning of the Vital Events Registration and Management System (VERMS) for the South Gondar Zone. The initiative was driven by the critical need to address the profound inefficiencies of the existing manual, paper-based system, which was characterized by extensive delays, high error rates, and significant vulnerability to data loss.

Through a methodical process, the project first established a clear understanding of the problem domain by analyzing the existing system's workflows, organizational structure, and technological limitations. This analysis translated into a detailed Software Requirements Specification (SRS) that defined the system's functional and non-functional needs. Subsequently, a robust system architecture was designed, proposing a secure, scalable, and offline-first web application built on a modern LAMP (Linux, Apache, MySQL, PHP) stack with a HTML5, CSS3, and JAVASCRIPT frontend, specifically chosen to overcome the challenge of unreliable internet connectivity in remote Kebeles.

The conclusion drawn from this endeavor is that the development and implementation of VERMS are not only technically feasible but also strategically imperative. The system is designed to directly eliminate the core pain points: it will reduce vital event registration processing time from weeks to hours, ensure data integrity through automated validation, and provide real-time access to information across all administrative levels. By enhancing security, privacy, and transparency, VERMS promises to restore citizen trust in a fundamental government service. Therefore, this project concludes that VERMS provides a viable, complete, and effective blueprint for digital transformation that will significantly improve governance, operational efficiency, and service delivery within the South Gondar Zone.

6.2 Recommendation

Based on the conclusive findings of this project, the following actionable recommendations are made to ensure the successful implementation and adoption of VERMS:

1. **Adopt a Phased Pilot Implementation Strategy:** The Zone Administration should not attempt a full zone-wide roll-out immediately. It is strongly recommended to begin with a pilot program in 3-5 strategically selected Kebele and one Woreda office. This approach will allow for real-world testing,

user training, and the identification of unforeseen challenges on a manageable scale, minimizing risk and building confidence before expanding.

2. **Commit to Essential Initial Investment:** For the pilot phase, a minimum initial investment is required for critical infrastructure:
 - a. **Hardware:** Procure reliable laptop computers and portable printers for the pilot Kebele offices.
 - b. **Connectivity:** Provide USB internet dongles or subsidize mobile data plans to ensure even minimal connectivity for data synchronization.
 - c. **Training:** Allocate a dedicated budget for developing comprehensive training materials in Amharic and conducting hands-on workshops for all staff involved in the pilot.
3. **Establish a Dedicated Support and Governance Structure:** A VERMS Support Team must be established from the onset, consisting of at least one IT staff member from the Zone office and the development team lead. This team will be responsible for providing immediate technical support during the pilot, managing user accounts, and overseeing the system's operation. A steering committee comprising management should meet weekly to review progress.
4. **Implement a Focused Data Migration Policy:** Avoid the costly and time-consuming effort of digitizing historical paper archives initially. The focus should be on new registrations. The recommendation is to declare a "Digital First" date, after which all new vital events must be registered exclusively through VERMS. Historical records can be digitized incrementally as resources allow.
5. **Initiate a Change Management and Awareness Campaign:** proactively address resistance to change by communicating the benefits of VERMS to both staff and citizens. This can be achieved through meetings, simple posters in offices, and demonstrations to showcase the speed and convenience of the new system.

6.3 Future Work

The current VERMS design provides a solid foundation for the core registration process. However, to further enhance its value and impact, the following future enhancements are proposed for consideration after the successful deployment and stabilization of the initial system:

1. **SMS/Email Notification System:** Integrate with an SMS gateway (e.g., Ethio Telecom's API) to automatically notify citizens about their application status

(e.g., "Your birth certificate is ready for collection at your Kebele office").

This would drastically reduce the number of unnecessary visits citizens make.

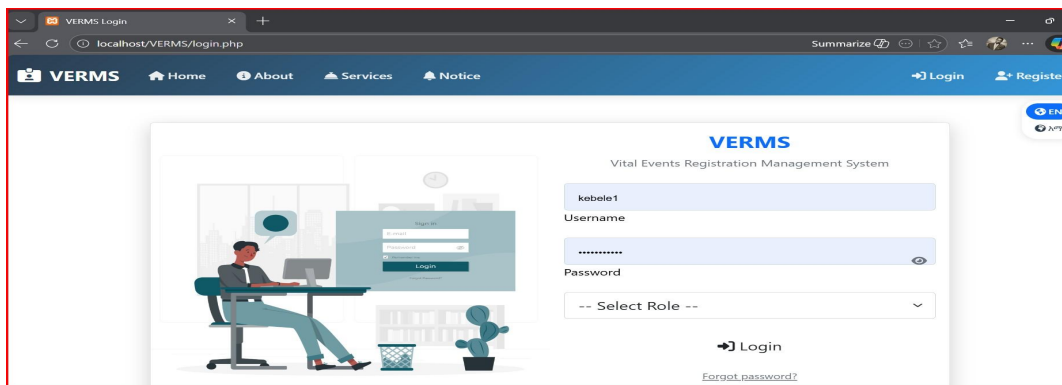
2. **Mobile Field Application Extension:** Develop a lightweight Android app version of VERMS for "Mobile Registrars." This app would be optimized for offline use on tablets, enabling teams to conduct registration drives in remote and pastoral communities, significantly increasing coverage.
3. **Advanced Business Intelligence Dashboard:** Upgrade the reporting module to a live, interactive dashboard with data visualization tools (e.g., integrated with Power BI or Meta-base). This would allow administrators to visualize trends, such as disease outbreaks mapped against death records or school enrollment projections based on birth rates, in real-time.
4. **System Integration with National Platforms:** Pursue API integration's with other government systems, most notably the National ID Database for automated citizen verification and the Health Management Information System (HMIS) to receive automatic notifications of births and deaths from health centers, reducing duplicate data entry and improving completeness.
5. **Block-chain-Based Verification Layer:** Explore the implementation of a block-chain to store hashes of issued certificates. This would create an immutable, decentralized public ledger for verification, making certificates virtually impossible to forge and greatly enhancing trust and security.

REFERENCE

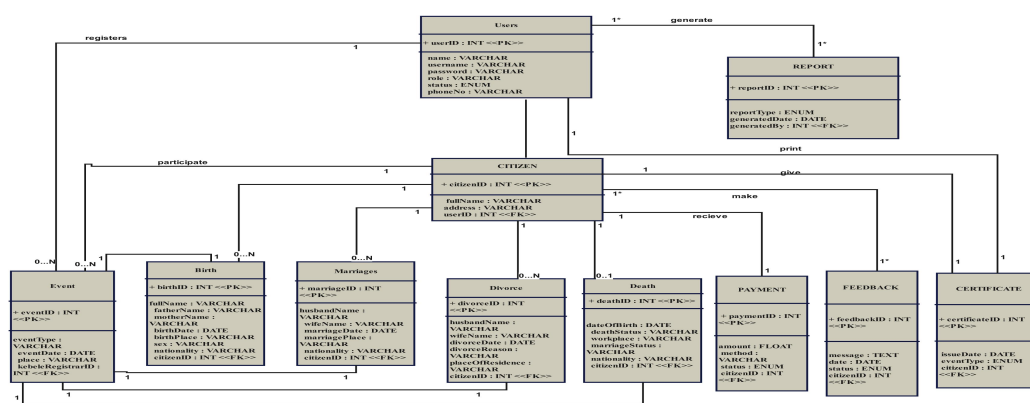
- [1] M. o. I. a. T. Federal Democratic Republic of Ethiopia, 2025. [Online]. Available: <https://www.lawethiopia.com>.
- [2] A.Union,2020. [Online]. Available: <https://au.int/en/documents/20200518/digital-transformation-strategy-africa-2020-2030>.
- [3] U. N. S. Division, Principles and Recommendations for a Vital Statistics System, United Nations, Dept. of Economic and Social Affairs, 2024.
- [4] E. V. E. R. A. (VERA), "National Vital Events Registration and Statistics Strategy 2016–2020," Government of Ethiopia, 2016.
- [5] I. Sommerville, Software Engineering (10th ed.), Pearson Education, 2016.
- [6] R. Elmasri and S. B. Navathe, Fundamentals of Database Systems, 7th ed., Pearson, 2016.

APPENDICES

Appendices A: System Screenshots Login Forms



Appendices B: Persistent ER Diagram



Appendices C:Source Code Samples

```
db_schema.sql - Visual Studio Code
db_schema.sql
1  -- Users table
2  CREATE TABLE users (
3      id INT AUTO_INCREMENT PRIMARY KEY,
4      fullname VARCHAR(100) NOT NULL,
5      username VARCHAR(50) NOT NULL UNIQUE,
6      email VARCHAR(100) NOT NULL UNIQUE,
7      password VARCHAR(255) NOT NULL,
8      role ENUM('kebele', 'woreda', 'zone', 'admin', 'citizen') NOT NULL,
9      created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
10 );
11
12 -- Birth Events
13 CREATE TABLE birth_events (
14     id INT AUTO_INCREMENT PRIMARY KEY,
15     user_id INT,
16     child_name VARCHAR(100),
17     date_of_birth DATE,
18     place_of_birth VARCHAR(100),
19     parent_names VARCHAR(200),
20     certificate_number VARCHAR(50),
21     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
22     FOREIGN KEY (user_id) REFERENCES users(id)
23 );
24
25 -- Marriage Events
26 CREATE TABLE marriage_events (
27     id INT AUTO_INCREMENT PRIMARY KEY,
28     user_id INT,
29     spouse1_name VARCHAR(100),
30     spouse2_name VARCHAR(100),
31     date_of_marriage DATE,
32     place_of_marriage VARCHAR(100),
```

```
db_schema.sql - Vital Events - Visual Studio Code

docs > db_schema.sql
39 CREATE TABLE death_events (
40     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
41     FOREIGN KEY (user_id) REFERENCES users(id)
42 );
43
44 -- Divorce Events
45 CREATE TABLE divorce_events (
46     id INT AUTO_INCREMENT PRIMARY KEY,
47     user_id INT,
48     spouse1_name VARCHAR(100),
49     spouse2_name VARCHAR(100),
50     date_of_divorce DATE,
51     place_of_divorce VARCHAR(100),
52     certificate_number VARCHAR(50),
53     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
54     FOREIGN KEY (user_id) REFERENCES users(id)
55 );
56
57 -- Feedback table
58 CREATE TABLE feedback (
59     id INT AUTO_INCREMENT PRIMARY KEY,
60     user_id INT,
61     message TEXT NOT NULL,
62     submitted_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
63     FOREIGN KEY (user_id) REFERENCES users(id)
64 );
65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100
101
102
103
104
105
106
107
108
109
110
111
112
113
114
115
116
117
118
119
120
121
122
123
124
125
126
127
128
129
130
131
132
133
134
135
136
137
138
139
140
141
142
143
144
145
146
147
148
149
150
151
152
153
154
155
156
157
158
159
160
161
162
163
164
165
166
167
168
169
170
171
172
173
174
175
176
177
178
179
180
181
182
183
184
185
186
187
188
189
190
191
192
193
194
195
196
197
198
199
200
201
202
203
204
205
206
207
208
209
210
211
212
213
214
215
216
217
218
219
220
221
222
223
224
225
226
227
228
229
230
231
232
233
234
235
236
237
238
239
240
241
242
243
244
245
246
247
248
249
250
251
252
253
254
255
256
257
258
259
260
261
262
263
264
265
266
267
268
269
270
271
272
273
274
275
276
277
278
279
280
281
282
283
284
285
286
287
288
289
290
291
292
293
294
295
296
297
298
299
300
301
302
303
304
305
306
307
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
327
328
329
330
331
332
333
334
335
336
337
338
339
340
341
342
343
344
345
346
347
348
349
350
351
352
353
354
355
356
357
358
359
360
361
362
363
364
365
366
367
368
369
370
371
372
373
374
375
376
377
378
379
380
381
382
383
384
385
386
387
388
389
390
391
392
393
394
395
396
397
398
399
400
401
402
403
404
405
406
407
408
409
410
411
412
413
414
415
416
417
418
419
420
421
422
423
424
425
426
427
428
429
430
431
432
433
434
435
436
437
438
439
440
441
442
443
444
445
446
447
448
449
450
451
452
453
454
455
456
457
458
459
460
461
462
463
464
465
466
467
468
469
470
471
472
473
474
475
476
477
478
479
480
481
482
483
484
485
486
487
488
489
490
491
492
493
494
495
496
497
498
499
500
501
502
503
504
505
506
507
508
509
510
511
512
513
514
515
516
517
518
519
520
521
522
523
524
525
526
527
528
529
530
531
532
533
534
535
536
537
538
539
540
541
542
543
544
545
546
547
548
549
550
551
552
553
554
555
556
557
558
559
560
561
562
563
564
565
566
567
568
569
570
571
572
573
574
575
576
577
578
579
580
581
582
583
584
585
586
587
588
589
590
591
592
593
594
595
596
597
598
599
600
601
602
603
604
605
606
607
608
609
610
611
612
613
614
615
616
617
618
619
620
621
622
623
624
625
626
627
628
629
630
631
632
633
634
635
636
637
638
639
640
641
642
643
644
645
646
647
648
649
650
651
652
653
654
655
656
657
658
659
660
661
662
663
664
665
666
667
668
669
670
671
672
673
674
675
676
677
678
679
680
681
682
683
684
685
686
687
688
689
690
691
692
693
694
695
696
697
698
699
700
701
702
703
704
705
706
707
708
709
710
711
712
713
714
715
716
717
718
719
720
721
722
723
724
725
726
727
728
729
730
731
732
733
734
735
736
737
738
739
740
741
742
743
744
745
746
747
748
749
750
751
752
753
754
755
756
757
758
759
760
761
762
763
764
765
766
767
768
769
770
771
772
773
774
775
776
777
778
779
780
781
782
783
784
785
786
787
788
789
790
791
792
793
794
795
796
797
798
799
800
801
802
803
804
805
806
807
808
809
810
811
812
813
814
815
816
817
818
819
820
821
822
823
824
825
826
827
828
829
830
831
832
833
834
835
836
837
838
839
840
841
842
843
844
845
846
847
848
849
850
851
852
853
854
855
856
857
858
859
860
861
862
863
864
865
866
867
868
869
870
871
872
873
874
875
876
877
878
879
880
881
882
883
884
885
886
887
888
889
890
891
892
893
894
895
896
897
898
899
900
901
902
903
904
905
906
907
908
909
910
911
912
913
914
915
916
917
918
919
920
921
922
923
924
925
926
927
928
929
930
931
932
933
934
935
936
937
938
939
940
941
942
943
944
945
946
947
948
949
950
951
952
953
954
955
956
957
958
959
960
961
962
963
964
965
966
967
968
969
970
971
972
973
974
975
976
977
978
979
980
981
982
983
984
985
986
987
988
989
990
991
992
993
994
995
996
997
998
999
1000
1001
1002
1003
1004
1005
1006
1007
1008
1009
1010
1011
1012
1013
1014
1015
1016
1017
1018
1019
1020
1021
1022
1023
1024
1025
1026
1027
1028
1029
1030
1031
1032
1033
1034
1035
1036
1037
1038
1039
1040
1041
1042
1043
1044
1045
1046
1047
1048
1049
1050
1051
1052
1053
1054
1055
1056
1057
1058
1059
1060
1061
1062
1063
1064
1065
1066
1067
1068
1069
1070
1071
1072
1073
1074
1075
1076
1077
1078
1079
1080
1081
1082
1083
1084
1085
1086
1087
1088
1089
1090
1091
1092
1093
1094
1095
1096
1097
1098
1099
1100
1101
1102
1103
1104
1105
1106
1107
1108
1109
1110
1111
1112
1113
1114
1115
1116
1117
1118
1119
1120
1121
1122
1123
1124
1125
1126
1127
1128
1129
1130
1131
1132
1133
1134
1135
1136
1137
1138
1139
1140
1141
1142
1143
1144
1145
1146
1147
1148
1149
1150
1151
1152
1153
1154
1155
1156
1157
1158
1159
1160
1161
1162
1163
1164
1165
1166
1167
1168
1169
1170
1171
1172
1173
1174
1175
1176
1177
1178
1179
1180
1181
1182
1183
1184
1185
1186
1187
1188
1189
1190
1191
1192
1193
1194
1195
1196
1197
1198
1199
1200
1201
1202
1203
1204
1205
1206
1207
1208
1209
1210
1211
1212
1213
1214
1215
1216
1217
1218
1219
1220
1221
1222
1223
1224
1225
1226
1227
1228
1229
1230
1231
1232
1233
1234
1235
1236
1237
1238
1239
1240
1241
1242
1243
1244
1245
1246
1247
1248
1249
1250
1251
1252
1253
1254
1255
1256
1257
1258
1259
1260
1261
1262
1263
1264
1265
1266
1267
1268
1269
1270
1271
1272
1273
1274
1275
1276
1277
1278
1279
1280
1281
1282
1283
1284
1285
1286
1287
1288
1289
1290
1291
1292
1293
1294
1295
1296
1297
1298
1299
1300
1301
1302
1303
1304
1305
1306
1307
1308
1309
1310
1311
1312
1313
1314
1315
1316
1317
1318
1319
1320
1321
1322
1323
1324
1325
1326
1327
1328
1329
1330
1331
1332
1333
1334
1335
1336
1337
1338
1339
1340
1341
1342
1343
1344
1345
1346
1347
1348
1349
1350
1351
1352
1353
1354
1355
1356
1357
1358
1359
1360
1361
1362
1363
1364
1365
1366
1367
1368
1369
1370
1371
1372
1373
1374
1375
1376
1377
1378
1379
1380
1381
1382
1383
1384
1385
1386
1387
1388
1389
1390
1391
1392
1393
1394
1395
1396
1397
1398
1399
1400
1401
1402
1403
1404
1405
1406
1407
1408
1409
1410
1411
1412
1413
1414
1415
1416
1417
1418
1419
1420
1421
1422
1423
1424
1425
1426
1427
1428
1429
1430
1431
1432
1433
1434
1435
1436
1437
1438
1439
1440
1441
1442
1443
1444
1445
1446
1447
1448
1449
1450
1451
1452
1453
1454
1455
1456
1457
1458
1459
1460
1461
1462
1463
1464
1465
1466
1467
1468
1469
1470
1471
1472
1473
1474
1475
1476
1477
1478
1479
1480
1481
1482
1483
1484
1485
1486
1487
1488
1489
1490
1491
1492
1493
1494
1495
1496
1497
1498
1499
1500
1501
1502
1503
1504
1505
1506
1507
1508
1509
1510
1511
1512
1513
1514
1515
1516
1517
1518
1519
1520
1521
1522
1523
1524
1525
1526
1527
1528
1529
1530
1531
1532
1533
1534
1535
1536
1537
1538
1539
1540
1541
1542
1543
1544
1545
1546
1547
1548
1549
1550
1551
1552
1553
1554
1555
1556
1557
1558
1559
1560
1561
1562
1563
1564
1565
1566
1567
1568
1569
1570
1571
1572
1573
1574
1575
1576
1577
1578
1579
1580
1581
1582
1583
1584
1585
1586
1587
1588
1589
1590
1591
1592
1593
1594
1595
1596
1597
1598
1599
1600
1601
1602
1603
1604
1605
1606
1607
1608
1609
1610
1611
1612
1613
1614
1615
1616
1617
1618
1619
1620
1621
1622
1623
1624
1625
1626
1627
1628
1629
1630
1631
1632
1633
1634
1635
1636
1637
1638
1639
1640
1641
1642
1643
1644
1645
1646
1647
1648
1649
1650
1651
1652
1653
1654
1655
1656
1657
1658
1659
1660
1661
1662
1663
1664
1665
1666
1667
1668
1669
1670
1671
1672
1673
1674
1675
1676
1677
1678
1679
1680
1681
1682
1683
1684
1685
1686
1687
1688
1689
1690
1691
1692
1693
1694
1695
1696
1697
1698
1699
1700
1701
1702
1703
1704
1705
1706
1707
1708
1709
1710
1711
1712
1713
1714
1715
1716
1717
1718
1719
1720
1721
1722
1723
1724
1725
1726
1727
1728
1729
1730
1731
1732
1733
1734
1735
1736
1737
1738
1739
1740
1741
1742
1743
1744
1745
1746
1747
1748
1749
1750
1751
1752
1753
1754
1755
1756
1757
1758
1759
1760
1761
1762
1763
1764
1765
1766
1767
1768
1769
1770
1771
1772
1773
1774
1775
1776
1777
1778
1779
1780
1781
1782
1783
1784
1785
1786
1787
1788
1789
1790
1791
1792
1793
1794
1795
1796
1797
1798
1799
1800
1801
1802
1803
1804
1805
1806
1807
1808
1809
1810
1811
1812
1813
1814
1815
1816
1817
1818
1819
1820
1821
1822
1823
1824
1825
1826
1827
1828
1829
1830
1831
1832
1833
1834
1835
1836
1837
1838
1839
1840
1841
1842
1843
1844
1845
1846
1847
1848
1849
1850
1851
1852
1853
1854
1855
1856
1857
1858
1859
1860
1861
1862
1863
1864
1865
1866
1867
1868
1869
1870
1871
1872
1873
1874
1875
1876
1877
1878
1879
1880
1881
1882
1883
1884
1885
1886
1887
1888
1889
1890
1891
1892
1893
1894
1895
1896
1897
1898
1899
1900
1901
1902
1903
1904
1905
1906
1907
1908
1909
1910
1911
1912
1913
1914
1915
1916
1917
1918
1919
1920
1921
1922
1923
1924
1925
1926
1927
1928
1929
1930
1931
1932
1933
1934
1935
1936
1937
1938
1939
1940
1941
1942
1943
1944
1945
1946
1947
1948
1949
1950
1951
1952
1953
1954
1955
1956
1957
1958
1959
1960
1961
1962
1963
1964
1965
1966
1967
1968
1969
1970
1971
1972
1973
1974
1975
1976
1977
1978
1979
1980
1981
1982
1983
1984
1985
1986
1987
1988
1989
1990
1991
1992
1993
1994
1995
1996
1997
1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
2027
2028
2029
2030
2031
2032
2033
2034
2035
2036
2037
2038
2039
2040
2041
2042
2043
2044
2045
2046
2047
2048
2049
2050
2051
2052
2053
2054
2055
2056
2057
2058
2059
2060
2061
2062
2063
2064
2065
2066
2067
2068
2069
2070
2071
2072
2073
2074
2075
2076
2077
2078
2079
2080
2081
2082
2083
2084
2085
2086
2087
2088
2089
2090
2091
2092
2093
2094
2095
2096
2097
2098
2099
2100
2101
2102
2103
2104
2105
2106
2107
2108
2109
2110
2111
2112
2113
2114
2115
2116
2117
2118
2119
2120
2121
2122
2123
2124
2125
2126
2127
2128
2129
2130
2131
2132
2133
2134
2135
2136
2137
2138
2139
2140
2141
2142
2143
2144
2145
2146
2147
2148
2149
2150
2151
2152
2153
2154
2155
2156
2157
2158
2159
2160
2161
2162
2163
2164
2165
2166
2167
2168
2169
2170
2171
2172
2173
2174
2175
2176
2177
2178
2179
2180
2181
2182
2183
2184
2185
2186
2187
2188
2189
2190
2191
2192
2193
2194
2195
2196
2197
2198
2199
2200
2201
2202
2203
2204
2205
2206
2207
2208
2209
2210
2211
2212
2213
2214
2215
2216
2217
2218
2219
2220
2221
2222
2223
2224
2225
2226
2227
2228
2229
2230
2231
2232
2233
2234
2235
2236
2237
2238
2239
2240
2241
2242
2243
2244
2245
2246
2247
2248
2249
2250
2251
2252
2253
2254
2255
2256
2257
2258
2259
2260
2261
2262
2263
2264
2265
2266
2267
2268
2269
2270
2271
2272
2273
2274
2275
2276
2277
2278
2279
2280
2281
2282
2283
2284
2285
2286
2287
2288
2289
2290
2291
2292
2293
2294
2295
2296
2297
2298
2299
2300
2301
2302
2303
2304
2305
2306
2307
2308
2309
2310
2311
2312
2313
2314
2315
2316
2317
2318
2319
2320
2321
2322
2323
2324
2325
2326
2327
2328
2329
2330
2331
2332
2333
2334
2335
2336
2337
2338
2339
2340
2341
2342
2343
2344
2345
2346
2347
2348
2349
2350
2351
2352
2353
2354
2355
2356
2357
2358
2359
2360
2361
2362
2363
2364
2365
2366
2367
2368
2369
2370
2371
2372
2373
2374
2375
2376
2377
2378
2379
2380
2381
2382
2383
2384
2385
2386
2387
2388
2389
2390
2391
2392
2393
2394
2395
2396
2397
2398
2399
2400
2401
2402
2403
2404
2405
2406
2407
2408
2409
2410
2411
2412
2413
2414
2415
2416
2417
2418
2419
2420
2421
2422
2423
2424
2425
2426
2427
2428
2429
2430
2431
2432
2433
2434
2435
2436
2437
2438
2439
2440
2441
2442
2443
2444
2445
2446
2447
2448
2449
2450
2451
2452
2453
2454
2455
2456
2457
2458
2459
2460
2461
2462
2463
2464
2465
2466
2467
2468
2469
2470
2471
2472
2473
2474
2475
2476
2477
2478
2479
2480
2481
2482
2483
2484
2485
2486
2487
2488
2489
2490
2491
2492
2493
2494
2495
2496
2497
2498
2499
2500
2501
2502
2503
2504
2505
2506
2507
2508
2509
2510
2511
2512
2513
2514
2515
2516
2517
2518
2519
2520
2521
2522
2523
2524
2525
2526
2527
2528
2529
2530
2531
2532
2533
2534
2535
2536
2537
2538
2539
2540
2541
2542
2543
2544
2545
2546
2547
2548
2549
2550
2551
2552
2553
2554
2555
2556
2557
2558
2559
2560
2561
2562
2563
2564
2565
2566
2567
2568
2569
2570
2571
2572
2573
2574
2575
2576
2577
2578
2579
2580
2581
2582
2583
2584
2585
2586
2587
2588
2589
2590
2591
2592
2593
2594
2595
2596
2597
2598
2599
2600
2601
2602
2603
2604
2605
2606
2607
2608
2609
2610
2611
2612
2613
2614
2615
2616
2617
2618
2619
2620
2621
2622
2623
2624
2625
2
```